

Table 1: Detailed results XH dataset by [Pessoa et al. \(2018\)](#): Minimum and average error gap to BKS per instance and solution method. Bolt marks the best result for each instance, underlined entries denote the best average performance. Blank cells indicate no feasible solution was found within the time limit by the given solution method.
Benchmarks: AILS by [Máximo et al. \(2022\)](#) and PyVRP by [Wouda et al. \(2024\)](#).

	Minimum Error Gap				Mean Error Gap			
	AILS	DRSCI	GSPI	PyVRP	AILS	DRSCI	GSPI	PyVRP
X101-FSMFD	0.0004	0.0009	0.0328	0.0	<u>0.0004</u>	0.0024	0.0363	0.0020
X106-FSMD	0.0028	0.0029	0.0330	0.0005	<u>0.0028</u>	0.0034	0.0486	<u>0.0013</u>
X110-HD	0.002	0.0156	0.0402	0.0249	<u>0.002</u>	0.0183	0.0544	0.0279
X115-HVRP	0.0	0.0123	-	0.0011	<u>0.0</u>	0.0300	-	<u>0.0013</u>
X120-FSMF	0.0	0.0256	0.0494	15.5474	<u>0.0</u>	0.0320	0.0538	16.0676
X125-HVRP	0.0015	0.0154	0.0138	-	<u>0.0015</u>	0.0197	0.0183	-
X129-FSMFD	0.0037	0.0190	0.0321	5.6310	<u>0.0037</u>	0.0225	0.0366	5.7698
X134-FSMD	0.0014	0.0026	0.0327	-0.0007	<u>0.0014</u>	0.0056	0.0419	0.0014
X139-HD	0.0037	0.0678	0.0602	0.0515	<u>0.0037</u>	0.0777	0.1004	0.0538
X143-FSMF	-0.0552	-0.0577	0.0310	-0.0585	<u>-0.0552</u>	-0.0521	0.0411	<u>-0.0562</u>
X148-HVRP	0.004	0.0263	0.0318	-	<u>0.004</u>	0.0303	0.0351	-
X153-FSMFD	0.0015	0.0043	0.0285	0.0048	<u>0.0015</u>	0.0064	0.0411	0.0064
X157-HD	0.0	0.0266	0.0533	-	<u>0.0</u>	0.0484	0.0570	-
X162-FSMD	0.0014	0.0023	0.0367	0.0	<u>0.0014</u>	0.0043	0.0453	0.0021
X167-FSMF	-0.0093	0.0088	0.0201	0.3305	<u>-0.0093</u>	0.0193	0.0306	11.6258
X172-HVRP	0.0027	0.0107	0.0461	0.0303	<u>0.0027</u>	0.0118	0.0503	0.0324
X176-FSMFD	0.0019	0.0365	0.0431	-	<u>0.0019</u>	0.0412	0.0680	-
X181-HD	0.0027	0.0345	0.0361	-	<u>0.0027</u>	0.0410	0.0399	-
X186-FSMD	0.0006	0.0047	0.0477	0.0087	<u>0.0006</u>	0.0065	0.0562	0.0132
X190-FSMF	0.0008	0.0152	0.0322	0.0312	<u>0.0008</u>	0.0219	0.0420	0.0564
X195-FSMF	-0.0201	-0.0157	0.0104	-0.0158	<u>-0.0201</u>	-0.0130	0.0132	-0.0132
X200-HD	0.0032	0.0132	0.0157	0.0032	<u>0.0032</u>	0.0169	0.0177	0.0039
X204-FSMD	-0.0021	0.0022	0.0411	0.0183	<u>-0.0021</u>	0.0066	0.0493	0.0285
X209-FSMFD	0.0015	0.0181	0.0404	0.0309	<u>0.0015</u>	0.0232	0.0530	0.0345
X214-HVRP	-0.0032	0.0263	0.0771	-0.0062	<u>-0.0032</u>	0.0413	0.0891	-0.0015
X219-HD	0.0042	0.0039	0.0049	-	<u>0.0042</u>	0.0047	0.0064	-
X223-HVRP	0.003	0.0169	0.0309	-	<u>0.003</u>	0.0332	0.0467	-
X228-FSMFD	0.0035	0.0142	0.0397	0.0201	<u>0.0035</u>	0.0175	0.0485	0.0221
X233-FSMD	0.0018	0.0078	0.0497	0.0014	<u>0.0018</u>	0.0137	0.0663	0.0158
X237-FSMF	-0.0322	-0.0055	0.0524	14.8823	<u>-0.0322</u>	0.0085	0.0682	14.9189
X242-FSMFD	0.005	0.0143	0.0257	3.5777	<u>0.005</u>	0.0153	0.0276	3.7333
X247-HVRP	0.0017	0.0203	0.0380	0.0109	<u>0.0017</u>	0.0229	0.0474	0.0161
X251-FSMD	0.0022	0.0067	0.0468	0.0171	<u>0.0022</u>	0.0080	0.0502	0.0177
X256-FSMF	-0.0091	-0.0068	0.0160	-0.0136	<u>-0.0091</u>	0.0002	0.0224	<u>-0.0104</u>
X261-HD	-0.0015	0.0245	0.0518	-0.0061	<u>-0.0015</u>	0.0343	0.0642	<u>-0.002</u>
X266-HD	0.0057	0.0194	0.0330	0.0699	<u>0.0057</u>	0.0261	0.0450	0.0789
X270-FSMD	0.0162	0.0058	0.0292	0.0156	<u>0.0162</u>	<u>0.006</u>	0.0366	0.0228
X275-HVRP	0.0046	0.0546	0.0533	-	<u>0.0046</u>	0.0584	0.0586	-
X280-FSMF	-0.0136	0.0171	0.0249	14.0816	<u>-0.0136</u>	0.0284	0.0392	14.2440
X284-FSMFD	0.0138	0.0230	0.0299	0.0336	<u>0.0138</u>	0.0304	0.0400	0.0373
X289-HVRP	0.0079	0.0160	0.0312	0.0295	<u>0.0079</u>	0.0229	0.0335	0.0311
X294-HD	0.0116	0.0159	0.0428	0.005	0.0116	0.0208	0.0497	<u>0.0081</u>
X298-FSMD	0.0018	0.0035	0.0362	0.0188	<u>0.0018</u>	0.0041	0.0485	0.0211
X303-FSMFD	0.0007	0.0004	0.0241	-0.0065	0.0007	0.0023	0.0431	<u>-0.0029</u>
X308-FSMF	-0.0388	-0.0006	0.0118	0.2924	<u>-0.0388</u>	0.0079	0.0231	0.3288
X313-FSMD	0.0290	0.0017	0.0179	0.0043	0.0290	<u>0.0034</u>	0.0193	0.0059
X317-HVRP	0.0014	0.0083	0.0098	-	<u>0.0014</u>	0.0095	0.0106	-
X322-HD	0.0155	0.0132	0.0415	0.0037	0.0155	0.0202	0.0451	<u>0.0059</u>
X327-FSMFD	0.0079	0.0241	0.0380	0.0387	<u>0.0079</u>	0.0270	0.0430	0.0405
X331-FSMF	-0.024	-0.0041	0.0219	19.6873	<u>-0.024</u>	0.0083	0.0286	20.1363
X336-FSMF	0.0036	0.0144	0.0371	0.1172	<u>0.0036</u>	0.0186	0.0412	0.1284
X344-FSMD	0.0026	0.0019	0.0396	0.0199	<u>0.0026</u>	0.0035	0.0444	0.0207
X351-HVRP	0.0028	0.0307	0.0316	0.0043	<u>0.0028</u>	0.0328	0.0345	0.0064
X359-HD	0.0025	0.0515	0.0528	0.0578	<u>0.0025</u>	0.0579	0.0618	0.0617
X367-FSMFD	-0.0116	0.0110	0.0261	0.0613	<u>-0.0116</u>	0.0174	0.0312	0.0660
X376-HD	0.0035	0.0078	0.0115	-	<u>0.0035</u>	0.0105	0.0129	-
X384-FSMF	0.0039	0.0082	0.0344	0.0046	<u>0.0039</u>	0.0093	0.0398	0.0084
X393-HVRP	0.0078	0.0320	0.0335	-	<u>0.0078</u>	0.0350	0.0361	-
X401-FSMFD	-0.0056	-0.0096	0.0159	-0.0131	<u>-0.0056</u>	-0.0082	0.0228	<u>-0.0126</u>
X411-FSMD	-0.0116	-0.0048	0.0703	-0.0153	<u>-0.0116</u>	0.0010	0.0850	-0.0092
X420-FSMD	0.005	0.0133	0.0353	-	<u>0.005</u>	0.0148	0.0564	-
X429-HVRP	0.0107	0.0209	0.0338	0.0037	0.0107	0.0239	0.0385	<u>0.0046</u>
X439-FSMF	0.001	0.0264	0.0490	10.3613	<u>0.001</u>	0.0294	0.0515	10.5406
X449-FSMFD	-0.0066	-0.0096	0.0231	-0.0148	<u>-0.0066</u>	-0.0080	0.0273	<u>-0.0142</u>
X459-HD	0.0093	0.0434	0.0634	0.0036	0.0093	0.0539	0.0662	<u>0.0064</u>
X469-HD	0.0086	0.0092	0.0561	0.0119	<u>0.0086</u>	0.0104	0.0592	0.0139
X480-FSMD	0.0076	0.0121	0.0368	5.1672	<u>0.0076</u>	0.0146	0.0396	5.2428
X491-FSMF	0.0057	0.0073	0.0348	0.0069	<u>0.0057</u>	0.0104	0.0378	0.0184

Table 1: (continued)

	Minimum Error Gap				Mean Error Gap			
	AILS	DRSCI	GSPI	PyVRP	AILS	DRSCI	GSPI	PyVRP
X502-FSMFD	-0.0155	-0.0074	0.0209	12.1135	<u>-0.0155</u>	-0.0008	0.0223	12.2447
X513-HVRP	-0.0003	0.0384	0.0431	0.0190	<u>-0.0003</u>	0.0427	0.0518	0.0245
X524-HD	0.0261	0.0286	0.0454	0.0468	<u>0.0261</u>	0.0325	0.0482	0.0551
X536-FSMFD	0.0062	0.0039	0.0702	0.0037	0.0062	0.0069	0.0736	<u>0.0047</u>
X548-FSMF	0.0025	0.0224	0.0432	11.2333	<u>0.0025</u>	0.0260	0.0448	11.3470
X561-FSMD	0.0034	0.0088	0.0388	0.0261	<u>0.0034</u>	0.0123	0.0487	0.0272
X573-HVRP	-0.0001	0.0180	0.0169	-	<u>-0.0001</u>	0.0233	0.0226	-
X586-FSMF	0.0022	0.0294	0.0525	3.0280	<u>0.0022</u>	0.0325	0.0546	3.0893
X599-FSMD	0.1222	0.0008	0.0496	0.0072	0.1222	<u>0.0009</u>	0.0520	0.0084
X613-HD	0.0143	0.0421	0.0413	0.0161	<u>0.0143</u>	0.0452	0.0480	0.0201
X627-HVRP	-0.0012	0.0165	0.0172	-	<u>-0.0012</u>	0.0210	0.0209	-
X641-FSMFD	0.0047	0.0012	0.0222	0.0085	0.0047	<u>0.0043</u>	0.0285	0.0173
X655-HD	0.0049	0.0248	0.0237	-	<u>0.0049</u>	0.0253	0.0252	-
X670-FSMF	0.0031	0.0175	0.0461	0.2692	<u>0.0031</u>	0.0236	0.0516	-
X685-FSMD	0.0420	0.0032	0.0472	0.0108	0.0420	<u>0.0058</u>	0.0546	0.0126
X701-HVRP	0.0043	0.0201	0.0171	-	<u>0.0043</u>	0.0211	0.0194	-
X716-FSMFD	-0.0106	-0.0035	0.0162	-0.0022	<u>-0.0106</u>	-0.0002	0.0215	0.0023
X733-FSMFD	0.0056	0.0157	0.0446	3.4048	<u>0.0056</u>	0.0183	0.0467	3.4998
X749-FSMF	0.0047	0.0100	0.0254	0.0066	<u>0.0047</u>	0.0109	0.0273	0.0082
X766-FSMD	0.0068	0.0089	0.0378	0.0208	<u>0.0068</u>	0.0105	0.0402	0.0236
X783-HD	0.0262	0.0416	0.0519	0.0196	<u>0.0262</u>	0.0508	0.0567	0.0263
X801-HVRP	0.0056	0.0441	0.0426	-	<u>0.0056</u>	0.0466	0.0442	-
X819-FSMD	0.0033	0.0079	0.0353	0.1571	<u>0.0033</u>	0.0095	0.0386	0.1667
X837-HD	0.0135	0.0200	0.0173	-	<u>0.0135</u>	0.0287	0.0218	-
X856-HVRP	0.0052	0.0308	0.0292	-	<u>0.0052</u>	0.0314	0.0311	-
X876-FSMF	-0.0928	-0.0841	-0.0632	-0.0905	<u>-0.0928</u>	-0.0821	-0.0583	-0.0844
X895-FSMFD	0.0040	-0.0025	0.0355	-0.0071	0.0040	-0.0008	0.0393	<u>-0.0046</u>
X916-FSMFD	0.0036	0.0264	0.0394	3.4529	<u>0.0036</u>	0.0279	0.0410	3.4807
X936-FSMD	0.0185	0.007	0.0363	0.0112	0.0185	<u>0.0088</u>	0.0389	0.0121
X957-HD	0.0115	0.0735	0.0775	-	<u>0.0115</u>	0.0857	0.0791	-
X979-HVRP	-0.0069	0.0097	0.0072	-0.0060	<u>-0.0069</u>	0.0133	0.0095	-0.0035
X1001-FSMF	-0.0558	-0.0417	0.0020	-0.0554	<u>-0.0558</u>	-0.0367	0.0082	-0.0533

Table 2: Detailed results FSMVRPTW dataset by Bräysy et al. (2009): Minimum and average error gap to BKS per instance and solution method. Bolt marks the best result for each instance, underlined entries denote the best average performance. Blank cells indicate no feasible solution was found within the time limit by the given solution method. Benchmark: PyVRP by Wouda et al. (2024).

	Minimum Error Gap			Mean Error Gap		
	DRSCI	GSPI	PyVRP	DRSCI	GSPI	PyVRP
c1.10.10_fsm_A	-0.0223	-0.0204	0.0069	<u>-0.0194</u>	-0.0175	0.0092
c1.10.10_fsm_C	-0.0472	-0.0406	-0.0379	<u>-0.0422</u>	-0.0402	-0.0325
c1.10.1_fsm_A	0.0006	0.0119	0.0876	<u>0.0012</u>	0.0125	0.0946
c1.10.1_fsm_C	0.0	0.0010	0.0153	<u>0.0001</u>	0.0012	0.0167
c1.10.2_fsm_A	-0.015	0.0017	0.0317	<u>-0.0126</u>	0.0059	0.0451
c1.10.2_fsm_C	-0.0246	-0.0206	-0.0125	<u>-0.0224</u>	-0.0161	-0.0096
c1.10.3_fsm_A	-0.0257	-0.0225	-0.0012	<u>-0.0211</u>	-0.0089	0.0055
c1.10.3_fsm_C	-0.0462	-0.0417	-0.0311	<u>-0.0429</u>	-0.0373	-0.0249
c1.10.4_fsm_A	-0.014	0.0072	0.0003	<u>-0.011</u>	0.0124	0.0038
c1.10.4_fsm_C	-0.0586	-0.0158	-0.0435	<u>-0.054</u>	-0.0105	-0.0329
c1.10.5_fsm_A	-0.004	0.0080	0.0702	<u>-0.0032</u>	0.0092	0.0784
c1.10.5_fsm_C	-0.0005	0.0010	0.0133	<u>-0.0005</u>	0.0010	0.0164
c1.10.6_fsm_A	-0.0071	0.0056	0.0536	<u>-0.0062</u>	0.0061	0.0773
c1.10.6_fsm_C	-0.0016	-0.0003	0.0144	<u>-0.0016</u>	-0.0002	0.0160
c1.10.7_fsm_A	-0.0129	0.0007	0.0357	<u>-0.0111</u>	0.0007	0.0513
c1.10.7_fsm_C	-0.0027	-0.0017	0.0118	<u>-0.0025</u>	-0.0014	0.0155
c1.10.8_fsm_A	-0.0096	-0.0005	0.0367	<u>-0.0087</u>	0.0026	0.0507
c1.10.8_fsm_C	-0.0162	-0.0133	0.0030	<u>-0.0113</u>	-0.0107	0.0059
c1.10.9_fsm_A	-0.0152	-0.0061	0.0394	<u>-0.0107</u>	-0.0024	0.0440
c1.10.9_fsm_C	-0.0239	-0.0248	-0.0120	<u>-0.0223</u>	-0.0217	-0.0088
c1.2.10_fsm_A	-0.0053	0.0185	0.0031	<u>0.0036</u>	0.0224	0.0049
c1.2.10_fsm_C	-0.007	-0.0025	-0.0038	<u>-0.0047</u>	-0.0025	0.0043
c1.2.1_fsm_A	0.0037	0.0335	0.0246	<u>0.0073</u>	0.0335	0.0294
c1.2.1_fsm_C	0.0005	0.0043	0.0005	0.0011	0.0043	<u>0.001</u>
c1.2.2_fsm_A	0.0066	0.0264	0.0024	0.0088	0.0303	<u>0.0058</u>
c1.2.2_fsm_C	0.0007	0.0051	0.0086	<u>0.0023</u>	0.0124	0.0112
c1.2.3_fsm_A	-0.0097	0.0032	-0.0040	<u>-0.0028</u>	0.0061	0.0003
c1.2.3_fsm_C	-0.0064	-0.0056	-0.0031	<u>-0.0055</u>	-0.0051	-0.0011
c1.2.4_fsm_A	-0.0094	0.0082	-0.0046	<u>-0.0015</u>	0.0155	<u>-0.0032</u>
c1.2.4_fsm_C	-0.0099	-0.0059	-0.0050	<u>-0.0072</u>	-0.0026	-0.0023
c1.2.5_fsm_A	-0.0016	0.0279	0.0120	<u>0.007</u>	0.0279	0.0211
c1.2.5_fsm_C	0.0002	0.0027	0.0005	<u>0.0004</u>	0.0027	0.0012
c1.2.6_fsm_A	0.0051	0.0269	0.0059	<u>0.0073</u>	0.0269	0.0149
c1.2.6_fsm_C	0.0003	0.0025	0.0003	<u>0.0003</u>	0.0025	0.0010
c1.2.7_fsm_A	0.0016	0.0229	0.0005	<u>0.005</u>	0.0229	0.0096
c1.2.7_fsm_C	0.001	0.0043	0.0104	<u>0.0019</u>	0.0043	0.0184
c1.2.8_fsm_A	-0.0016	0.0277	-0.0036	0.0044	0.0277	<u>0.0035</u>
c1.2.8_fsm_C	-0.0001	0.0146	0.0064	<u>0.0024</u>	0.0146	0.0180
c1.2.9_fsm_A	-0.0075	0.0140	-0.0052	-0.0005	0.0140	<u>-0.0011</u>
c1.2.9_fsm_C	-0.0005	0.0027	0.0054	<u>0.0007</u>	0.0027	0.0095
c1.4.10_fsm_A	-0.0103	0.0008	-0.012	-0.0062	0.0072	<u>-0.0076</u>
c1.4.10_fsm_C	-0.0145	-0.0144	-0.0101	<u>-0.0129</u>	-0.0122	-0.0068
c1.4.1_fsm_A	0.0039	0.0142	0.0406	<u>0.0059</u>	0.0142	0.0501
c1.4.1_fsm_C	0.0002	0.0011	0.0008	<u>0.0005</u>	0.0011	0.0090
c1.4.2_fsm_A	-0.0036	0.0245	0.0002	<u>-0.0002</u>	0.0329	0.0066
c1.4.2_fsm_C	-0.0037	0.0132	-0.0001	<u>-0.0023</u>	0.0179	0.0034
c1.4.3_fsm_A	-0.0102	-0.0053	-0.0098	<u>-0.0076</u>	0.0038	-0.0032
c1.4.3_fsm_C	-0.0119	0.0114	-0.0056	<u>-0.0057</u>	0.0207	-0.0018
c1.4.4_fsm_A	-0.0115	-0.0006	-0.0070	<u>-0.0081</u>	0.0048	0.0007
c1.4.4_fsm_C	-0.0224	-0.0099	-0.0229	-0.0193	0.0004	<u>-0.0208</u>
c1.4.5_fsm_A	-0.0048	0.0081	0.0372	<u>-0.0032</u>	0.0081	0.0384
c1.4.5_fsm_C	0.0	0.0018	0.0054	<u>0.0004</u>	0.0018	0.0124
c1.4.6_fsm_A	-0.0142	0.0017	0.0126	<u>-0.0117</u>	0.0017	0.0255
c1.4.6_fsm_C	0.0002	0.0027	0.0098	<u>0.0008</u>	0.0027	0.0139
c1.4.7_fsm_A	-0.0148	0.0005	0.0076	<u>-0.0132</u>	0.0005	0.0209
c1.4.7_fsm_C	0.0	0.0034	0.0138	<u>0.0005</u>	0.0034	0.0168
c1.4.8_fsm_A	-0.0114	0.0065	-0.0023	<u>-0.0086</u>	0.0072	0.0053
c1.4.8_fsm_C	-0.005	0.0004	0.0036	<u>-0.0044</u>	0.0011	0.0075
c1.4.9_fsm_A	-0.0132	0.0030	-0.0007	<u>-0.0104</u>	0.0073	0.0025
c1.4.9_fsm_C	-0.016	-0.0141	-0.0125	<u>-0.0144</u>	-0.0127	-0.0069
c1.6.10_fsm_A	-0.0156	-0.0084	-0.0167	<u>-0.0107</u>	-0.0048	-0.0070
c1.6.10_fsm_C	-0.0086	-0.0078	-0.0051	<u>-0.0038</u>	-0.0019	-0.0026
c1.6.1_fsm_A	0.0034	0.0206	0.0612	<u>0.0045</u>	0.0230	0.0660
c1.6.1_fsm_C	-0.0004	0.0035	0.0079	<u>-0.0</u>	0.0037	0.0125
c1.6.2_fsm_A	-0.0002	0.0216	0.0159	<u>0.0013</u>	0.0236	0.0223
c1.6.2_fsm_C	-0.0037	-0.0030	0.0031	<u>-0.0028</u>	0.0030	0.0066
c1.6.3_fsm_A	-0.0088	-0.0004	-0.0033	<u>-0.0059</u>	0.0067	0.0038
c1.6.3_fsm_C	-0.0263	-0.0131	-0.0235	<u>-0.0237</u>	-0.0099	-0.0107

Table 2: (continued)

	Minimum Error Gap			Mean Error Gap		
	DRSCI	GSPI	PyVRP	DRSCI	GSPI	PyVRP
c1.6.4_fsm_A	-0.0064	0.0143	-0.0054	<u>-0.0012</u>	0.0174	-0.0000
c1.6.4_fsm_C	-0.0237	0.0046	-0.0200	<u>-0.0222</u>	0.0097	-0.0158
c1.6.5_fsm_A	-0.0043	0.0176	0.0332	<u>-0.0024</u>	0.0191	0.0390
c1.6.5_fsm_C	0.0001	0.0037	0.0137	<u>0.0006</u>	0.0038	0.0181
c1.6.6_fsm_A	-0.0102	0.0109	0.0327	<u>-0.009</u>	0.0117	0.0408
c1.6.6_fsm_C	-0.0008	0.0034	0.0095	<u>-0.0005</u>	0.0034	0.0144
c1.6.7_fsm_A	-0.0112	0.0101	0.0334	<u>-0.0091</u>	0.0123	0.0459
c1.6.7_fsm_C	-0.0006	0.0031	0.0198	<u>-0.0002</u>	0.0031	0.0216
c1.6.8_fsm_A	-0.0081	0.0062	0.0361	<u>-0.0054</u>	0.0108	0.0422
c1.6.8_fsm_C	-0.0041	0.0008	0.0076	<u>-0.003</u>	0.0023	0.0099
c1.6.9_fsm_A	-0.0134	-0.0005	0.0076	<u>-0.0099</u>	0.0039	0.0159
c1.6.9_fsm_C	-0.0121	-0.0044	-0.0026	<u>-0.0098</u>	-0.0037	0.0008
c1.8.10_fsm_A	-0.0207	-0.0179	-0.0045	<u>-0.0152</u>	-0.0134	-0.0005
c1.8.10_fsm_C	-0.0348	-0.0327	-0.0273	<u>-0.0284</u>	<u>-0.0298</u>	-0.0227
c1.8.1_fsm_A	0.0021	0.0142	0.0871	<u>0.0031</u>	0.0165	0.0917
c1.8.1_fsm_C	-0.0005	0.0008	0.0148	<u>-0.0004</u>	0.0012	0.0178
c1.8.2_fsm_A	-0.004	0.0163	0.0345	<u>-0.0025</u>	0.0192	0.0403
c1.8.2_fsm_C	-0.0091	-0.0027	0.0056	<u>-0.0079</u>	-0.0022	0.0080
c1.8.3_fsm_A	-0.017	-0.0016	-0.0016	<u>-0.0137</u>	0.0064	0.0026
c1.8.3_fsm_C	-0.0336	-0.0167	-0.0161	<u>-0.0311</u>	-0.0071	-0.0102
c1.8.4_fsm_A	-0.0175	0.0056	-0.0035	<u>-0.0144</u>	0.0068	-0.0029
c1.8.4_fsm_C	-0.0432	-0.0238	-0.0401	<u>-0.0409</u>	-0.0165	-0.0306
c1.8.5_fsm_A	-0.0047	0.0106	0.0483	<u>-0.0042</u>	0.0115	0.0614
c1.8.5_fsm_C	-0.0002	0.0011	0.0170	<u>-0.0</u>	0.0011	0.0197
c1.8.6_fsm_A	-0.0122	0.0032	0.0377	<u>-0.0114</u>	0.0040	0.0596
c1.8.6_fsm_C	-0.0016	-0.0000	0.0178	<u>-0.0015</u>	0.0002	0.0199
c1.8.7_fsm_A	-0.0116	0.0042	0.0379	<u>-0.01</u>	0.0045	0.0560
c1.8.7_fsm_C	-0.0017	0.0003	0.0163	<u>-0.0014</u>	0.0003	0.0183
c1.8.8_fsm_A	-0.0138	-0.0027	0.0387	<u>-0.0103</u>	0.0015	0.0432
c1.8.8_fsm_C	-0.0133	-0.0116	0.0028	<u>-0.0083</u>	<u>-0.011</u>	0.0055
c1.8.9_fsm_A	-0.0236	-0.0114	0.0160	<u>-0.0196</u>	-0.0052	0.0217
c1.8.9_fsm_C	-0.0266	-0.0255	-0.0191	<u>-0.0254</u>	-0.0251	-0.0154
c2.10.10_fsm_A	-0.0419	-0.0417	-0.0258	<u>-0.0356</u>	<u>-0.0362</u>	-0.0120
c2.10.10_fsm_C	-0.0957	-0.098	-0.0879	<u>-0.0951</u>	-0.0937	-0.0847
c2.10.1_fsm_A	-0.0392	-0.0314	-0.0151	<u>-0.0366</u>	-0.0314	0.0209
c2.10.1_fsm_C	-0.0755	-0.0719	-0.0647	<u>-0.0739</u>	-0.0719	-0.0500
c2.10.2_fsm_A	-0.0577	-0.0393	-0.0036	<u>-0.0516</u>	-0.0355	0.0035
c2.10.2_fsm_C	-0.0786	-0.0705	-0.0656	<u>-0.0762</u>	-0.0695	-0.0542
c2.10.3_fsm_A	-0.0454	-0.0286	-0.0141	<u>-0.0377</u>	-0.0219	-0.0043
c2.10.3_fsm_C	-0.0935	-0.0879	-0.0854	<u>-0.0923</u>	-0.0871	-0.0793
c2.10.4_fsm_A	-0.0281	-0.0270	-0.0249	<u>-0.0239</u>	-0.0210	-0.0146
c2.10.4_fsm_C	-0.0874	-0.0817	-0.0799	<u>-0.0812</u>	-0.0771	-0.0703
c2.10.5_fsm_A	-0.0485	-0.0383	0.0124	<u>-0.045</u>	-0.0322	0.0252
c2.10.5_fsm_C	-0.0848	-0.0831	-0.0674	<u>-0.0832</u>	-0.0818	-0.0588
c2.10.6_fsm_A	-0.0516	-0.0386	0.0078	<u>-0.0465</u>	-0.0350	0.0201
c2.10.6_fsm_C	-0.103	-0.0998	-0.0903	<u>-0.0999</u>	-0.0976	-0.0823
c2.10.7_fsm_A	-0.044	-0.0331	0.0041	<u>-0.0407</u>	-0.0290	0.0145
c2.10.7_fsm_C	-0.0876	-0.0849	-0.0726	<u>-0.0852</u>	-0.0845	-0.0672
c2.10.8_fsm_A	-0.0488	-0.0249	-0.0033	<u>-0.0422</u>	-0.0212	0.0099
c2.10.8_fsm_C	-0.0935	-0.0880	-0.0858	<u>-0.0926</u>	-0.0869	-0.0813
c2.10.9_fsm_A	-0.0499	-0.0334	-0.0201	<u>-0.0432</u>	-0.0287	-0.0086
c2.10.9_fsm_C	-0.0984	-0.0916	-0.0857	<u>-0.0962</u>	-0.0910	-0.0812
c2.2.10_fsm_A	0.0055	0.0354	0.0005	0.0154	0.0485	<u>0.012</u>
c2.2.10_fsm_C	-0.0097	0.0075	-0.0105	0.0026	0.0075	<u>-0.004</u>
c2.2.1_fsm_A	-0.0067	0.0193	-0.0044	0.0107	0.0193	<u>0.0099</u>
c2.2.1_fsm_C	0.0020	0.0059	0.0002	<u>0.0057</u>	0.0059	0.0108
c2.2.2_fsm_A	-0.0315	0.0052	-0.0120	<u>-0.004</u>	0.0052	0.0095
c2.2.2_fsm_C	-0.0128	-0.0051	-0.015	<u>-0.0066</u>	-0.0051	-0.0016
c2.2.3_fsm_A	0.0119	0.0361	0.0140	0.0223	0.0362	<u>0.015</u>
c2.2.3_fsm_C	-0.0183	-0.0116	-0.0170	<u>-0.0156</u>	-0.0116	-0.0120
c2.2.4_fsm_A	-0.0108	0.0235	-0.0128	0.0007	0.0303	<u>-0.0058</u>
c2.2.4_fsm_C	-0.0262	-0.0145	-0.0213	<u>-0.022</u>	-0.0145	-0.0143
c2.2.5_fsm_A	-0.0177	0.0227	0.0034	<u>0.0059</u>	0.0266	0.0167
c2.2.5_fsm_C	-0.0049	0.0045	-0.012	-0.0001	0.0047	<u>-0.0085</u>
c2.2.6_fsm_A	0.0076	0.0206	0.0096	0.0135	0.0206	<u>0.0122</u>
c2.2.6_fsm_C	0.0001	0.0001	-0.0122	0.0001	0.0001	<u>-0.0017</u>
c2.2.7_fsm_A	-0.0107	0.0143	-0.0115	<u>0.0001</u>	0.0143	0.0015
c2.2.7_fsm_C	-0.0147	0.0009	-0.0114	-0.0040	0.0009	<u>-0.0051</u>
c2.2.8_fsm_A	0.0008	0.0270	0.0094	<u>0.0144</u>	0.0271	0.0254
c2.2.8_fsm_C	-0.0066	-0.0066	-0.0205	-0.0066	-0.0064	<u>-0.0096</u>
c2.2.9_fsm_A	-0.0077	0.0104	0.0017	<u>-0.0015</u>	0.0235	0.0075
c2.2.9_fsm_C	-0.0076	0.0030	-0.0134	<u>-0.005</u>	0.0031	-0.0027

Table 2: (continued)

	Minimum Error Gap			Mean Error Gap		
	DRSCI	GSPI	PyVRP	DRSCI	GSPI	PyVRP
c2.4.10_fsm_A	-0.0345	-0.0173	-0.0269	<u>-0.0251</u>	-0.0002	-0.0078
c2.4.10_fsm_C	-0.0454	-0.0366	-0.0456	<u>-0.0359</u>	-0.0321	-0.0353
c2.4.1_fsm_A	-0.0339	-0.0192	-0.0107	<u>-0.0204</u>	-0.0192	0.0112
c2.4.1_fsm_C	-0.0257	-0.0257	-0.0222	<u>-0.0225</u>	<u>-0.0257</u>	-0.0088
c2.4.2_fsm_A	-0.0019	0.0058	0.0134	<u>0.0021</u>	0.0158	0.0320
c2.4.2_fsm_C	-0.0370	-0.0337	-0.0389	<u>-0.0353</u>	-0.0330	-0.0286
c2.4.3_fsm_A	-0.0249	-0.0082	-0.0230	<u>-0.0212</u>	-0.0005	-0.0099
c2.4.3_fsm_C	-0.037	-0.0259	-0.0160	<u>-0.0302</u>	-0.0168	-0.0135
c2.4.4_fsm_A	-0.0291	-0.0036	-0.0204	<u>-0.0094</u>	0.0029	<u>-0.0165</u>
c2.4.4_fsm_C	-0.0466	-0.0284	-0.0447	<u>-0.0399</u>	-0.0241	-0.0344
c2.4.5_fsm_A	-0.027	-0.0090	0.0014	<u>-0.0244</u>	-0.0067	0.0118
c2.4.5_fsm_C	-0.0340	-0.0308	-0.0399	<u>-0.0325</u>	-0.0305	-0.0249
c2.4.6_fsm_A	-0.0222	0.0058	-0.0048	<u>-0.0067</u>	0.0061	0.0060
c2.4.6_fsm_C	-0.0259	-0.0146	-0.0343	<u>-0.0189</u>	-0.0144	-0.0125
c2.4.7_fsm_A	-0.0362	-0.0021	-0.0176	<u>-0.0271</u>	0.0002	0.0009
c2.4.7_fsm_C	-0.0430	-0.0394	-0.05	<u>-0.0401</u>	-0.0390	<u>-0.0432</u>
c2.4.8_fsm_A	-0.0445	-0.0112	-0.0074	<u>-0.0317</u>	-0.0044	0.0035
c2.4.8_fsm_C	-0.0517	-0.0501	-0.0541	<u>-0.0492</u>	-0.0462	-0.0329
c2.4.9_fsm_A	-0.0341	-0.0124	-0.0207	<u>-0.024</u>	-0.0038	-0.0097
c2.4.9_fsm_C	-0.0321	-0.0226	-0.0229	<u>-0.0258</u>	-0.0224	-0.0170
c2.6.10_fsm_A	-0.0305	-0.0104	-0.0247	<u>-0.024</u>	-0.0046	-0.0151
c2.6.10_fsm_C	-0.0705	-0.0608	-0.0649	<u>-0.0657</u>	-0.0586	-0.0615
c2.6.1_fsm_A	-0.0426	-0.0309	-0.0102	<u>-0.0341</u>	-0.0309	0.0017
c2.6.1_fsm_C	-0.0365	-0.0322	-0.0173	<u>-0.0322</u>	-0.0322	-0.0002
c2.6.2_fsm_A	-0.0482	-0.0274	-0.0401	<u>-0.0468</u>	-0.0260	-0.0277
c2.6.2_fsm_C	-0.0504	-0.0462	-0.0447	<u>-0.0483</u>	-0.0449	-0.0326
c2.6.3_fsm_A	-0.0289	-0.0115	-0.0200	<u>-0.0248</u>	-0.0061	-0.0118
c2.6.3_fsm_C	-0.0616	-0.0526	-0.0586	<u>-0.0584</u>	-0.0507	-0.0491
c2.6.4_fsm_A	-0.0342	-0.0182	-0.0306	<u>-0.0267</u>	-0.0117	-0.0206
c2.6.4_fsm_C	-0.0747	-0.0625	-0.0613	<u>-0.0656</u>	-0.0587	-0.0525
c2.6.5_fsm_A	-0.0435	-0.0316	-0.0120	<u>-0.0375</u>	-0.0276	0.0064
c2.6.5_fsm_C	-0.059	-0.0549	-0.0476	<u>-0.0529</u>	<u>-0.0549</u>	-0.0360
c2.6.6_fsm_A	-0.0359	-0.0180	-0.0209	<u>-0.0349</u>	-0.0139	-0.0032
c2.6.6_fsm_C	-0.0625	-0.0586	-0.0634	<u>-0.0588</u>	-0.0583	-0.0588
c2.6.7_fsm_A	-0.0383	-0.0241	-0.0369	<u>-0.0291</u>	-0.0194	-0.0083
c2.6.7_fsm_C	-0.0707	-0.0650	-0.0656	<u>-0.0677</u>	-0.0647	-0.0507
c2.6.8_fsm_A	-0.0466	-0.0164	-0.0260	<u>-0.034</u>	-0.0086	-0.0122
c2.6.8_fsm_C	-0.0495	-0.0451	-0.0408	<u>-0.0473</u>	-0.0441	-0.0331
c2.6.9_fsm_A	-0.0365	-0.0152	-0.0184	<u>-0.0313</u>	-0.0074	-0.0104
c2.6.9_fsm_C	-0.0698	-0.0663	-0.0693	<u>-0.0672</u>	-0.0640	-0.0642
c2.8.10_fsm_A	-0.0304	-0.0233	-0.0115	<u>-0.0231</u>	-0.0197	-0.0024
c2.8.10_fsm_C	-0.0679	-0.0673	-0.0701	<u>-0.0592</u>	<u>-0.0658</u>	-0.0613
c2.8.1_fsm_A	-0.0449	-0.0282	0.0037	<u>-0.0417</u>	-0.0275	0.0216
c2.8.1_fsm_C	-0.0442	-0.0414	-0.0428	<u>-0.0431</u>	-0.0409	-0.0307
c2.8.2_fsm_A	-0.0412	-0.0254	-0.0062	<u>-0.0336</u>	-0.0162	0.0020
c2.8.2_fsm_C	-0.0734	-0.0632	-0.0626	<u>-0.0718</u>	-0.0619	-0.0550
c2.8.3_fsm_A	-0.0357	-0.0271	-0.0278	<u>-0.0341</u>	-0.0246	-0.0224
c2.8.3_fsm_C	-0.0762	-0.0698	-0.0728	<u>-0.0655</u>	<u>-0.0667</u>	-0.0602
c2.8.4_fsm_A	-0.0215	-0.0126	-0.0063	<u>-0.0172</u>	-0.0082	-0.0019
c2.8.4_fsm_C	-0.0742	-0.0699	-0.0766	<u>-0.0707</u>	-0.0667	-0.0663
c2.8.5_fsm_A	-0.0491	-0.0231	0.0208	<u>-0.039</u>	-0.0165	0.0244
c2.8.5_fsm_C	-0.0647	-0.0575	-0.0541	<u>-0.0636</u>	-0.0575	-0.0514
c2.8.6_fsm_A	-0.0508	-0.0264	-0.0090	<u>-0.0424</u>	-0.0183	0.0099
c2.8.6_fsm_C	-0.0623	-0.0585	-0.0520	<u>-0.0605</u>	-0.0560	-0.0448
c2.8.7_fsm_A	-0.042	-0.0281	-0.0078	<u>-0.036</u>	-0.0242	-0.0007
c2.8.7_fsm_C	0.0584	0.0629	0.0705	<u>0.0619</u>	0.0646	0.0773
c2.8.8_fsm_A	-0.0369	-0.0287	-0.0119	<u>-0.0326</u>	-0.0187	-0.0043
c2.8.8_fsm_C	-0.0774	-0.0743	-0.0681	<u>-0.0758</u>	-0.0695	-0.0649
c2.8.9_fsm_A	-0.0295	-0.0188	-0.0005	<u>-0.0221</u>	-0.0114	0.0052
c2.8.9_fsm_C	-0.091	-0.0877	-0.0823	<u>-0.0879</u>	-0.0859	-0.0754
r1.10.10_fsm_A	-0.0216	0.0706	-0.0202	<u>-0.0204</u>	0.0756	-0.0002
r1.10.10_fsm_C	-0.0347	0.0840	-0.0358	<u>-0.0271</u>	0.0918	-0.0065
r1.10.1_fsm_A	0.009	0.0210	0.0269	<u>0.0188</u>	0.0270	0.0389
r1.10.1_fsm_C	-0.02	-0.0024	0.0084	<u>-0.0101</u>	0.0109	0.0180
r1.10.2_fsm_A	-0.0063	0.0580	-0.0033	<u>0.0042</u>	0.0657	0.0183
r1.10.2_fsm_C	-0.035	0.0319	-0.0081	<u>-0.0172</u>	0.0578	0.0152
r1.10.3_fsm_A	-0.0178	0.0747	0.0082	<u>-0.0064</u>	0.0904	0.0130
r1.10.3_fsm_C	-0.0349	0.0813	-0.0416	<u>-0.0192</u>	0.0890	-0.0170
r1.10.4_fsm_A	-0.0222	0.0510	-0.0252	<u>-0.0154</u>	0.0650	<u>-0.0223</u>
r1.10.4_fsm_C	-0.0364	0.0640	-0.0363	<u>-0.0289</u>	0.0712	<u>-0.0325</u>
r1.10.5_fsm_A	-0.0188	0.0839	0.0378	<u>-0.0083</u>	0.0947	0.0642
r1.10.5_fsm_C	-0.0333	0.0634	-0.0052	<u>-0.0281</u>	0.0695	0.0103

Table 2: (continued)

	Minimum Error Gap			Mean Error Gap		
	DRSCI	GSPI	PyVRP	DRSCI	GSPI	PyVRP
r1.10.6_fsm_A	-0.0153	0.1017	0.0061	<u>-0.0103</u>	0.1062	0.0164
r1.10.6_fsm_C	-0.0421	0.0864	-0.0170	<u>-0.0346</u>	0.0956	-0.0115
r1.10.7_fsm_A	-0.0279	0.0675	-0.0241	<u>-0.0241</u>	0.0785	-0.0152
r1.10.7_fsm_C	-0.0389	0.0867	-0.0283	<u>-0.0339</u>	0.1015	-0.0211
r1.10.8_fsm_A	-0.0200	0.0462	-0.0288	<u>-0.0146</u>	0.0536	<u>-0.0242</u>
r1.10.8_fsm_C	-0.0331	0.0715	-0.039	<u>-0.0268</u>	0.0735	<u>-0.0345</u>
r1.10.9_fsm_A	-0.0247	0.0818	0.0027	<u>-0.016</u>	0.0929	0.0243
r1.10.9_fsm_C	-0.0297	0.0878	-0.0020	<u>-0.0245</u>	0.1045	0.0074
r1.2.10_fsm_A	-0.0078	0.0968	-0.0218	0.0021	0.1173	<u>-0.0213</u>
r1.2.10_fsm_C	-0.0087	0.1061	-0.0163	-0.0056	0.1212	<u>-0.0124</u>
r1.2.1_fsm_A	0.0153	0.0344	0.0029	0.0255	0.0378	<u>0.0047</u>
r1.2.1_fsm_C	-0.0025	0.0127	-0.0017	<u>-0.0013</u>	0.0133	-0.0002
r1.2.2_fsm_A	0.0134	0.0206	-0.0106	0.0257	0.0252	<u>-0.0062</u>
r1.2.2_fsm_C	-0.0071	0.0116	-0.0117	-0.0009	0.0127	<u>-0.0095</u>
r1.2.3_fsm_A	-0.0029	0.0509	-0.0049	0.0162	0.0613	<u>0.0046</u>
r1.2.3_fsm_C	-0.0116	0.0227	-0.0085	<u>-0.0108</u>	0.0287	-0.0078
r1.2.4_fsm_A	-0.0126	0.0664	-0.0166	0.0016	0.0805	<u>-0.0165</u>
r1.2.4_fsm_C	-0.0115	0.1034	-0.0172	-0.0042	0.1164	<u>-0.016</u>
r1.2.5_fsm_A	-0.0046	0.0079	-0.0102	0.0070	0.0224	<u>-0.0055</u>
r1.2.5_fsm_C	-0.0114	0.0093	-0.0116	-0.0086	0.0275	<u>-0.0106</u>
r1.2.6_fsm_A	0.0033	0.0595	-0.0026	<u>0.0095</u>	0.0672	0.0096
r1.2.6_fsm_C	-0.0147	0.0269	-0.0132	<u>-0.0085</u>	0.0360	-0.0077
r1.2.7_fsm_A	-0.0096	0.0938	-0.0246	<u>-0.0024</u>	0.1030	<u>-0.0242</u>
r1.2.7_fsm_C	-0.0247	0.0681	-0.0253	<u>-0.0188</u>	0.0901	-0.0155
r1.2.8_fsm_A	-0.0049	0.0656	-0.0128	0.0013	0.0744	<u>-0.012</u>
r1.2.8_fsm_C	-0.0234	0.0759	-0.0244	-0.0191	0.0896	<u>-0.0235</u>
r1.2.9_fsm_A	-0.0078	0.0549	-0.0196	-0.0051	0.0730	<u>-0.0112</u>
r1.2.9_fsm_C	-0.0219	0.0361	-0.0222	-0.0172	0.0415	<u>-0.0194</u>
r1.4.10_fsm_A	-0.0186	0.1013	-0.027	<u>-0.0164</u>	0.1124	-0.0155
r1.4.10_fsm_C	-0.0305	0.0987	-0.0251	<u>-0.0266</u>	0.1183	-0.0205
r1.4.1_fsm_A	0.0190	0.0459	0.0036	0.0255	0.0485	0.0109
r1.4.1_fsm_C	0.0008	0.0035	-0.0053	0.0053	0.0052	<u>-0.0031</u>
r1.4.2_fsm_A	-0.0034	0.0426	-0.0205	0.0151	0.0495	<u>-0.0033</u>
r1.4.2_fsm_C	-0.0209	0.0173	-0.0288	-0.0182	0.0269	<u>-0.0226</u>
r1.4.3_fsm_A	-0.0229	0.0780	-0.0284	-0.0085	0.0854	<u>-0.0136</u>
r1.4.3_fsm_C	-0.034	0.0744	-0.0333	<u>-0.0259</u>	0.0901	-0.0254
r1.4.4_fsm_A	-0.0127	0.0707	-0.02	-0.0112	0.0751	<u>-0.0185</u>
r1.4.4_fsm_C	-0.0266	0.0701	-0.0359	-0.0244	0.0899	<u>-0.0295</u>
r1.4.5_fsm_A	-0.0108	0.0355	0.0014	<u>-0.0058</u>	0.0508	0.0171
r1.4.5_fsm_C	-0.014	0.0378	-0.0117	<u>-0.0095</u>	0.0420	-0.0070
r1.4.6_fsm_A	-0.0212	0.0961	-0.0175	<u>-0.0167</u>	0.1065	0.0040
r1.4.6_fsm_C	-0.0344	0.0662	-0.0301	<u>-0.0273</u>	0.0756	-0.0186
r1.4.7_fsm_A	-0.0207	0.0610	-0.0246	-0.0178	0.0799	<u>-0.0239</u>
r1.4.7_fsm_C	-0.036	0.0889	-0.0350	<u>-0.0301</u>	0.1089	-0.0282
r1.4.8_fsm_A	-0.0139	0.0646	-0.0213	-0.0107	0.0691	<u>-0.018</u>
r1.4.8_fsm_C	-0.0291	0.0867	-0.0305	-0.0249	0.0971	<u>-0.0298</u>
r1.4.9_fsm_A	-0.0203	0.0839	-0.0077	<u>-0.0132</u>	0.0880	0.0044
r1.4.9_fsm_C	-0.016	0.0615	-0.0112	<u>-0.0116</u>	0.0706	-0.0015
r1.6.10_fsm_A	-0.0203	0.0736	-0.0083	<u>-0.0142</u>	0.0861	0.0112
r1.6.10_fsm_C	-0.0393	0.1045	-0.0368	<u>-0.0312</u>	0.1087	-0.0222
r1.6.1_fsm_A	0.0153	0.0522	0.0314	<u>0.0279</u>	0.0581	0.0406
r1.6.1_fsm_C	-0.0096	-0.0059	0.0015	-0.0025	<u>-0.0029</u>	0.0034
r1.6.2_fsm_A	0.0043	0.0646	0.0157	<u>0.0126</u>	0.0722	0.0265
r1.6.2_fsm_C	-0.0237	0.0313	-0.0207	<u>-0.0156</u>	0.0396	-0.0140
r1.6.3_fsm_A	-0.0122	0.0657	0.0100	<u>0.0014</u>	0.0778	0.0234
r1.6.3_fsm_C	-0.0276	0.0841	-0.0351	-0.0244	0.0896	<u>-0.0247</u>
r1.6.4_fsm_A	-0.0142	0.0683	-0.0304	-0.0100	0.0792	<u>-0.0295</u>
r1.6.4_fsm_C	-0.0236	0.0701	-0.0321	-0.0199	0.0914	<u>-0.0276</u>
r1.6.5_fsm_A	-0.0112	0.0679	0.0240	<u>-0.008</u>	0.0746	0.0344
r1.6.5_fsm_C	-0.0272	0.0163	-0.0059	<u>-0.0222</u>	0.0418	-0.0016
r1.6.6_fsm_A	-0.0232	0.0766	0.0027	<u>-0.0155</u>	0.0859	0.0185
r1.6.6_fsm_C	-0.0274	0.0863	-0.0284	<u>-0.0215</u>	0.0956	-0.0162
r1.6.7_fsm_A	-0.0262	0.0768	-0.0224	<u>-0.0191</u>	0.0935	-0.0048
r1.6.7_fsm_C	-0.026	0.1059	-0.0219	<u>-0.0211</u>	0.1181	-0.0130
r1.6.8_fsm_A	-0.0201	0.0559	-0.0271	-0.0177	0.0604	<u>-0.0249</u>
r1.6.8_fsm_C	-0.0299	0.0766	-0.0391	-0.0238	0.0892	<u>-0.0346</u>
r1.6.9_fsm_A	-0.0183	0.0923	0.0207	<u>-0.0124</u>	0.0992	0.0329
r1.6.9_fsm_C	-0.0244	0.0921	-0.0094	<u>-0.0185</u>	0.1018	-0.0007
r1.8.10_fsm_A	-0.0277	0.0612	0.0014	<u>-0.0235</u>	0.0673	0.0108
r1.8.10_fsm_C	-0.0394	0.0884	-0.0376	<u>-0.0352</u>	0.0946	-0.0103
r1.8.1_fsm_A	0.0015	0.0268	0.0081	<u>0.0076</u>	0.0311	0.0179
r1.8.1_fsm_C	-0.0213	-0.0133	-0.0026	<u>-0.0127</u>	0.0010	0.0057

Table 2: (continued)

	Minimum Error Gap			Mean Error Gap		
	DRSCI	GSPI	PyVRP	DRSCI	GSPI	PyVRP
r1.8.2_fsm_A	-0.0065	0.0311	-0.0039	0.0074	0.0435	<u>0.0049</u>
r1.8.2_fsm_C	-0.0216	0.0246	-0.0152	<u>-0.0134</u>	0.0340	-0.0092
r1.8.3_fsm_A	-0.01	0.0750	-0.0034	<u>0.0002</u>	0.0882	0.0058
r1.8.3_fsm_C	-0.0299	0.0846	-0.04	-0.0202	0.0955	<u>-0.0236</u>
r1.8.4_fsm_A	-0.0242	0.0467	-0.032	-0.0170	0.0644	<u>-0.0276</u>
r1.8.4_fsm_C	-0.0287	0.0863	-0.0315	-0.0218	0.0939	<u>-0.0279</u>
r1.8.5_fsm_A	-0.0149	0.0557	0.0333	<u>-0.009</u>	0.0758	0.0372
r1.8.5_fsm_C	-0.0356	0.0530	-0.0097	<u>-0.0285</u>	0.0689	-0.0032
r1.8.6_fsm_A	-0.0259	0.0836	0.0002	<u>-0.0186</u>	0.0945	0.0186
r1.8.6_fsm_C	-0.0269	0.1016	-0.0356	<u>-0.0213</u>	0.1104	-0.0059
r1.8.7_fsm_A	-0.0185	0.0731	-0.0039	<u>-0.0135</u>	0.0839	0.0015
r1.8.7_fsm_C	-0.0392	0.0893	-0.0273	<u>-0.0328</u>	0.0994	-0.0166
r1.8.8_fsm_A	-0.0247	0.0440	-0.0325	-0.0211	0.0577	-0.0308
r1.8.8_fsm_C	-0.0253	0.0817	-0.0341	-0.0231	0.0909	<u>-0.0308</u>
r1.8.9_fsm_A	-0.0214	0.0843	-0.0020	<u>-0.0166</u>	0.0934	0.0183
r1.8.9_fsm_C	-0.0318	0.0916	-0.0082	<u>-0.0266</u>	0.1081	0.0061
r2.10.10_fsm_A	-0.1776	-0.1535	-0.1716	<u>-0.1631</u>	-0.1323	-0.1524
r2.10.10_fsm_C	-0.0790	-0.0918	-0.0945	-0.0725	-0.0840	<u>-0.0879</u>
r2.10.1_fsm_A	-0.1079	-0.1078	-0.1192	-0.0937	-0.0966	<u>-0.1181</u>
r2.10.1_fsm_C	-0.0431	-0.0420	-0.0545	-0.0367	-0.0394	<u>-0.046</u>
r2.10.2_fsm_A	-0.1433	-0.1338	-0.1546	-0.1319	-0.1196	<u>-0.1447</u>
r2.10.2_fsm_C	-0.0389	-0.0405	-0.0553	-0.0279	-0.0300	<u>-0.0479</u>
r2.10.3_fsm_A	-0.1264	-0.1192	-0.151	-0.1199	-0.1098	<u>-0.1295</u>
r2.10.3_fsm_C	-0.0643	-0.0635	-0.0807	-0.0547	-0.0580	<u>-0.0715</u>
r2.10.4_fsm_A	-0.1160	-0.0404	-0.1233	-0.0986	-0.0232	<u>-0.118</u>
r2.10.4_fsm_C	-0.0990	-0.0447	-0.1078	-0.0911	-0.0331	<u>-0.1011</u>
r2.10.5_fsm_A	-0.1585	-0.1716	-0.1585	-0.1372	<u>-0.1584</u>	-0.1537
r2.10.5_fsm_C	-0.0545	-0.0521	-0.0626	-0.0480	-0.0502	<u>-0.0591</u>
r2.10.6_fsm_A	-0.1339	-0.1202	-0.1444	-0.1192	-0.1044	<u>-0.1297</u>
r2.10.6_fsm_C	-0.0675	-0.0741	-0.0806	-0.0579	-0.0662	<u>-0.0752</u>
r2.10.7_fsm_A	-0.1348	-0.0783	-0.1193	<u>-0.1206</u>	-0.0630	-0.1085
r2.10.7_fsm_C	-0.0749	-0.0858	-0.0988	-0.0642	-0.0747	<u>-0.093</u>
r2.10.8_fsm_A	-0.0991	-0.0345	-0.1139	-0.0782	-0.0274	<u>-0.1013</u>
r2.10.8_fsm_C	-0.0787	-0.0271	-0.1046	-0.0699	-0.0190	<u>-0.0939</u>
r2.10.9_fsm_A	-0.1727	-0.1761	-0.1591	-0.1604	<u>-0.1683</u>	-0.1493
r2.10.9_fsm_C	-0.0642	-0.0719	-0.0767	-0.0535	-0.0676	<u>-0.0722</u>
r2.2.10_fsm_A	-0.0398	-0.0433	-0.0436	-0.0263	-0.0268	<u>-0.0436</u>
r2.2.10_fsm_C	-0.0584	-0.0641	-0.0614	-0.0390	<u>-0.0626</u>	-0.0542
r2.2.1_fsm_A	-0.0267	0.0154	-0.0652	-0.0146	0.0218	<u>-0.0441</u>
r2.2.1_fsm_C	0.0069	0.0320	-0.0044	0.0153	0.0321	<u>0.0035</u>
r2.2.2_fsm_A	-0.0500	-0.0314	-0.078	-0.0351	-0.0232	<u>-0.0706</u>
r2.2.2_fsm_C	-0.0270	0.0085	-0.0377	-0.0156	0.0130	<u>-0.0292</u>
r2.2.3_fsm_A	-0.0151	0.0280	-0.0269	-0.0066	0.0424	<u>-0.0199</u>
r2.2.3_fsm_C	-0.0188	0.0181	-0.0362	-0.0091	0.0183	<u>-0.0305</u>
r2.2.4_fsm_A	-0.0178	-0.0277	-0.0287	-0.0118	0.0145	<u>-0.0251</u>
r2.2.4_fsm_C	-0.0602	-0.0518	-0.077	-0.0535	-0.0476	<u>-0.068</u>
r2.2.5_fsm_A	-0.0397	-0.0154	-0.068	-0.0322	-0.0123	<u>-0.0557</u>
r2.2.5_fsm_C	-0.0263	-0.0174	-0.0388	-0.0211	-0.0165	<u>-0.035</u>
r2.2.6_fsm_A	-0.0688	-0.0919	-0.0789	-0.0607	<u>-0.0737</u>	-0.0592
r2.2.6_fsm_C	-0.0350	-0.0188	-0.0448	-0.0255	-0.0153	<u>-0.0408</u>
r2.2.7_fsm_A	-0.0282	-0.0312	-0.0352	-0.0178	-0.0138	<u>-0.0217</u>
r2.2.7_fsm_C	-0.0341	-0.0159	-0.04	-0.0238	0.0050	<u>-0.0312</u>
r2.2.8_fsm_A	-0.0212	0.0276	-0.0306	-0.0157	0.0340	<u>-0.03</u>
r2.2.8_fsm_C	-0.0663	-0.0629	-0.0684	-0.0525	-0.0410	<u>-0.0592</u>
r2.2.9_fsm_A	-0.0875	-0.0894	-0.0827	-0.0681	<u>-0.0827</u>	-0.0611
r2.2.9_fsm_C	-0.0294	-0.0259	-0.0543	-0.0241	-0.0242	<u>-0.044</u>
r2.4.10_fsm_A	-0.0536	-0.0367	-0.0923	-0.0500	-0.0211	<u>-0.0745</u>
r2.4.10_fsm_C	-0.0671	-0.0825	-0.0839	-0.0587	-0.0776	<u>-0.0815</u>
r2.4.1_fsm_A	-0.0786	-0.0680	-0.1149	-0.0683	-0.0628	<u>-0.0925</u>
r2.4.1_fsm_C	-0.0269	0.0044	-0.0403	-0.0129	0.0150	<u>-0.0362</u>
r2.4.2_fsm_A	-0.0494	-0.0289	-0.0897	-0.0392	-0.0267	<u>-0.0752</u>
r2.4.2_fsm_C	-0.0244	0.0142	-0.0418	-0.0150	0.0225	<u>-0.0374</u>
r2.4.3_fsm_A	-0.0593	-0.0683	-0.0659	-0.0421	-0.0393	<u>-0.0574</u>
r2.4.3_fsm_C	-0.0529	-0.0303	-0.0664	-0.0377	-0.0239	<u>-0.0635</u>
r2.4.4_fsm_A	-0.0539	0.0199	-0.0583	<u>-0.0392</u>	0.0344	-0.0257
r2.4.4_fsm_C	-0.0486	-0.0350	-0.0772	-0.0421	-0.0202	<u>-0.0634</u>
r2.4.5_fsm_A	-0.0876	-0.1112	-0.1079	-0.0686	-0.0846	<u>-0.0961</u>
r2.4.5_fsm_C	-0.0455	-0.0432	-0.0704	-0.0385	-0.0397	<u>-0.0644</u>
r2.4.6_fsm_A	-0.0435	-0.0385	-0.083	-0.0305	-0.0244	<u>-0.047</u>
r2.4.6_fsm_C	-0.0585	-0.0487	-0.081	-0.0452	-0.0384	<u>-0.0737</u>
r2.4.7_fsm_A	-0.0416	-0.0057	-0.0742	-0.0345	0.0223	<u>-0.0573</u>
r2.4.7_fsm_C	-0.0631	-0.0502	-0.0815	-0.0553	-0.0374	<u>-0.0783</u>

Table 2: (continued)

	Minimum Error Gap			Mean Error Gap		
	DRSCI	GSPI	PyVRP	DRSCI	GSPI	PyVRP
r2.4.8_fsm_A	-0.0082	0.0538	-0.0344	0.0007	0.0864	<u>-0.0237</u>
r2.4.8_fsm_C	-0.0645	-0.0206	-0.0799	-0.0559	0.0046	<u>-0.0751</u>
r2.4.9_fsm_A	-0.0974	-0.0889	-0.126	-0.0765	-0.0804	<u>-0.1012</u>
r2.4.9_fsm_C	-0.0521	-0.0550	-0.0724	-0.0443	-0.0521	<u>-0.0697</u>
r2.6.10_fsm_A	-0.1188	-0.0645	-0.1212	-0.1023	-0.0391	<u>-0.1163</u>
r2.6.10_fsm_C	-0.0594	-0.0589	-0.0772	-0.0491	-0.0523	<u>-0.0747</u>
r2.6.1_fsm_A	-0.1190	-0.1222	-0.1466	-0.0996	-0.1119	<u>-0.1304</u>
r2.6.1_fsm_C	-0.0336	-0.0063	-0.0354	-0.0219	0.0002	<u>-0.0315</u>
r2.6.2_fsm_A	-0.1074	-0.1202	-0.1601	-0.0917	-0.1101	<u>-0.1297</u>
r2.6.2_fsm_C	-0.0308	-0.0191	-0.0412	-0.0189	-0.0139	<u>-0.0367</u>
r2.6.3_fsm_A	-0.0810	-0.0763	-0.0975	-0.0683	-0.0584	<u>-0.0828</u>
r2.6.3_fsm_C	-0.0491	-0.0375	-0.0627	-0.0386	-0.0348	<u>-0.0594</u>
r2.6.4_fsm_A	-0.0819	-0.0139	-0.0759	<u>-0.0654</u>	0.0146	<u>-0.0651</u>
r2.6.4_fsm_C	-0.0574	-0.0464	-0.0743	-0.0531	-0.0338	<u>-0.0724</u>
r2.6.5_fsm_A	-0.1417	-0.1627	-0.1499	-0.1210	<u>-0.1539</u>	<u>-0.1281</u>
r2.6.5_fsm_C	-0.0308	-0.0303	-0.049	-0.0242	-0.0253	<u>-0.0452</u>
r2.6.6_fsm_A	-0.0931	-0.0900	-0.1017	<u>-0.0808</u>	-0.0728	<u>-0.0805</u>
r2.6.6_fsm_C	-0.0521	-0.0609	-0.077	-0.0427	-0.0548	<u>-0.0738</u>
r2.6.7_fsm_A	-0.0930	-0.0558	-0.1129	<u>-0.0828</u>	-0.0265	<u>-0.0791</u>
r2.6.7_fsm_C	-0.0537	-0.0657	-0.0751	-0.0458	-0.0515	<u>-0.0716</u>
r2.6.8_fsm_A	-0.0649	-0.0147	-0.101	-0.0551	0.0009	<u>-0.0781</u>
r2.6.8_fsm_C	-0.0656	-0.0385	-0.0781	-0.0569	-0.0089	<u>-0.0691</u>
r2.6.9_fsm_A	-0.1404	-0.1259	-0.1412	-0.1078	-0.1075	<u>-0.1284</u>
r2.6.9_fsm_C	-0.0394	-0.0490	-0.0584	-0.0308	-0.0447	<u>-0.0552</u>
r2.8.10_fsm_A	-0.1497	-0.1051	-0.1649	-0.1388	-0.0917	<u>-0.1532</u>
r2.8.10_fsm_C	-0.0660	-0.0713	-0.0855	-0.0562	-0.0675	<u>-0.082</u>
r2.8.1_fsm_A	-0.1047	-0.1026	-0.1413	-0.0923	-0.0986	<u>-0.1393</u>
r2.8.1_fsm_C	-0.0379	-0.0387	-0.0493	-0.0303	-0.0347	<u>-0.0455</u>
r2.8.2_fsm_A	-0.1158	-0.1325	-0.1532	-0.1044	-0.1261	<u>-0.127</u>
r2.8.2_fsm_C	-0.0571	-0.0591	-0.0623	-0.0478	-0.0524	<u>-0.06</u>
r2.8.3_fsm_A	-0.1250	-0.1249	-0.1436	-0.1166	-0.1070	<u>-0.1375</u>
r2.8.3_fsm_C	-0.0616	-0.0679	-0.0726	-0.0506	-0.0590	<u>-0.0698</u>
r2.8.4_fsm_A	-0.1407	-0.0495	-0.1251	<u>-0.1317</u>	-0.0422	<u>-0.1222</u>
r2.8.4_fsm_C	-0.0765	-0.0733	-0.0938	-0.0735	-0.0512	<u>-0.0913</u>
r2.8.5_fsm_A	-0.1299	-0.1409	-0.1434	-0.1181	<u>-0.1276</u>	<u>-0.1266</u>
r2.8.5_fsm_C	-0.0492	-0.0543	-0.0682	-0.0423	-0.0499	<u>-0.0624</u>
r2.8.6_fsm_A	-0.137	-0.1116	-0.1353	<u>-0.1196</u>	-0.0978	<u>-0.1184</u>
r2.8.6_fsm_C	-0.0618	-0.0653	-0.0777	-0.0533	-0.0630	<u>-0.0731</u>
r2.8.7_fsm_A	-0.1469	-0.1305	-0.1713	-0.1440	-0.1051	<u>-0.1579</u>
r2.8.7_fsm_C	-0.0827	-0.0846	-0.0918	-0.0724	-0.0728	<u>-0.0888</u>
r2.8.8_fsm_A	-0.1169	-0.0369	-0.1279	<u>-0.1036</u>	-0.0304	<u>-0.1007</u>
r2.8.8_fsm_C	-0.0740	-0.0422	-0.0965	-0.0690	-0.0272	<u>-0.0886</u>
r2.8.9_fsm_A	-0.1316	-0.1371	-0.1541	-0.1155	<u>-0.1304</u>	<u>-0.1208</u>
r2.8.9_fsm_C	-0.0633	-0.0572	-0.0723	-0.0463	-0.0545	<u>-0.0696</u>
rc1.10.10_fsm_A	-0.0268	0.1039	-0.0268	<u>-0.0216</u>	0.1142	<u>-0.0142</u>
rc1.10.10_fsm_C	-0.0344	0.1079	-0.0411	<u>-0.0294</u>	0.1303	<u>-0.0236</u>
rc1.10.1_fsm_A	-0.0245	0.0229	0.0180	<u>-0.0108</u>	0.0337	<u>0.0240</u>
rc1.10.1_fsm_C	-0.0392	0.0131	-0.0193	<u>-0.0317</u>	0.0230	<u>-0.0137</u>
rc1.10.2_fsm_A	-0.031	0.0813	-0.0147	<u>-0.0186</u>	0.0887	<u>0.0064</u>
rc1.10.2_fsm_C	-0.0422	0.0625	-0.0231	<u>-0.036</u>	0.0792	<u>-0.0166</u>
rc1.10.3_fsm_A	-0.0275	0.0782	-0.0062	<u>-0.0236</u>	0.0910	<u>-0.0003</u>
rc1.10.3_fsm_C	-0.0359	0.0993	-0.0434	<u>-0.0309</u>	0.1083	<u>-0.0252</u>
rc1.10.4_fsm_A	-0.0205	0.0508	-0.0253	-0.0149	0.0624	<u>-0.0195</u>
rc1.10.4_fsm_C	-0.0301	0.0619	-0.0382	-0.0261	0.0794	<u>-0.0308</u>
rc1.10.5_fsm_A	-0.0343	0.0382	-0.0034	<u>-0.0186</u>	0.0567	<u>0.0092</u>
rc1.10.5_fsm_C	-0.0431	0.0419	-0.0485	<u>-0.0349</u>	0.0516	<u>-0.0229</u>
rc1.10.6_fsm_A	-0.024	0.0763	0.0138	<u>-0.0164</u>	0.0857	<u>0.0329</u>
rc1.10.6_fsm_C	-0.0498	0.0749	-0.0576	<u>-0.0379</u>	0.0862	<u>-0.0374</u>
rc1.10.7_fsm_A	-0.0252	0.0880	0.0197	<u>-0.0143</u>	0.0956	<u>0.0378</u>
rc1.10.7_fsm_C	-0.0476	0.0866	-0.0531	<u>-0.0414</u>	0.1041	<u>-0.0268</u>
rc1.10.8_fsm_A	-0.0212	0.1040	0.0005	<u>-0.0171</u>	0.1079	<u>0.0083</u>
rc1.10.8_fsm_C	-0.0349	0.1095	-0.0149	<u>-0.0272</u>	0.1308	<u>0.0018</u>
rc1.10.9_fsm_A	-0.0226	0.0964	-0.0102	<u>-0.0186</u>	0.1041	<u>0.0101</u>
rc1.10.9_fsm_C	-0.0388	0.1059	-0.0426	<u>-0.0333</u>	0.1172	<u>-0.0277</u>
rc1.2.10_fsm_A	-0.0048	0.0805	-0.0215	0.0002	0.1058	<u>-0.0194</u>
rc1.2.10_fsm_C	-0.0156	0.1214	-0.0177	-0.0131	0.1272	<u>-0.0166</u>
rc1.2.1_fsm_A	0.0060	0.0711	-0.0039	0.0155	0.0761	<u>-0.0049</u>
rc1.2.1_fsm_C	-0.0134	0.0140	-0.0128	-0.0096	0.0158	<u>-0.0119</u>
rc1.2.2_fsm_A	-0.0109	0.0752	-0.0116	-0.0027	0.0868	<u>-0.0068</u>
rc1.2.2_fsm_C	-0.0172	0.0803	-0.0261	-0.0157	0.0841	<u>-0.0175</u>
rc1.2.3_fsm_A	-0.0049	0.0897	-0.0199	0.0000	0.1006	<u>-0.0176</u>
rc1.2.3_fsm_C	-0.0268	0.0777	-0.0273	-0.0244	0.0886	<u>-0.0266</u>

Table 2: (continued)

	Minimum Error Gap			Mean Error Gap		
	DRSCI	GSPI	PyVRP	DRSCI	GSPI	PyVRP
rc1.2.4.fsm_A	-0.0138	0.0613	-0.0214	-0.0019	0.0719	<u>-0.0199</u>
rc1.2.4.fsm_C	-0.0175	0.0667	-0.019	-0.0146	0.0885	<u>-0.0189</u>
rc1.2.5.fsm_A	-0.0021	0.0256	-0.0295	0.0034	0.0511	<u>-0.0195</u>
rc1.2.5.fsm_C	-0.0165	0.0213	-0.0141	<u>-0.0108</u>	0.0382	<u>-0.0105</u>
rc1.2.6.fsm_A	-0.0107	0.0345	-0.0322	-0.0013	0.0485	<u>-0.0168</u>
rc1.2.6.fsm_C	-0.0149	0.0252	-0.0227	-0.0081	0.0469	<u>-0.0148</u>
rc1.2.7.fsm_A	-0.0134	0.0588	-0.027	-0.0043	0.0855	<u>-0.025</u>
rc1.2.7.fsm_C	-0.0165	0.0597	-0.0183	-0.0112	0.0664	<u>-0.014</u>
rc1.2.8.fsm_A	-0.0021	0.0988	-0.0183	0.0011	0.1133	<u>-0.0163</u>
rc1.2.8.fsm_C	-0.0317	0.0890	-0.0339	-0.0299	0.0924	<u>-0.0329</u>
rc1.2.9.fsm_A	-0.0206	0.0876	-0.0275	-0.0025	0.0964	<u>-0.0255</u>
rc1.2.9.fsm_C	-0.0276	0.0739	-0.0293	<u>-0.0254</u>	0.0845	<u>-0.0246</u>
rc1.4.10.fsm_A	-0.0125	0.1199	-0.0174	-0.0089	0.1261	<u>-0.0164</u>
rc1.4.10.fsm_C	-0.0381	0.1071	-0.0408	-0.0337	0.1330	<u>-0.0389</u>
rc1.4.1.fsm_A	-0.0046	0.0241	-0.006	<u>0.0019</u>	0.0355	0.0058
rc1.4.1.fsm_C	-0.0125	0.0312	-0.0103	<u>-0.0093</u>	0.0456	-0.0084
rc1.4.2.fsm_A	-0.0200	0.0705	-0.0264	<u>-0.0183</u>	0.0808	-0.0172
rc1.4.2.fsm_C	-0.0305	0.0396	-0.0299	<u>-0.0275</u>	0.0537	-0.0218
rc1.4.3.fsm_A	-0.0212	0.0775	-0.0287	-0.0170	0.0892	<u>-0.026</u>
rc1.4.3.fsm_C	-0.0352	0.0631	-0.0422	-0.0306	0.0991	<u>-0.0378</u>
rc1.4.4.fsm_A	-0.0172	0.0865	-0.0201	-0.0131	0.1024	<u>-0.0175</u>
rc1.4.4.fsm_C	-0.0294	0.1004	-0.0339	-0.0262	0.1127	<u>-0.0323</u>
rc1.4.5.fsm_A	-0.0094	0.0487	-0.0109	-0.0010	0.0634	<u>-0.0045</u>
rc1.4.5.fsm_C	-0.0218	0.0458	-0.0219	<u>-0.0209</u>	0.0490	<u>-0.0202</u>
rc1.4.6.fsm_A	-0.0122	0.0739	-0.0127	-0.0034	0.0810	<u>-0.009</u>
rc1.4.6.fsm_C	-0.0345	0.0347	-0.0372	<u>-0.031</u>	0.0518	-0.0302
rc1.4.7.fsm_A	-0.0129	0.0806	-0.0301	-0.0097	0.0897	<u>-0.0176</u>
rc1.4.7.fsm_C	-0.0322	0.0679	-0.0348	-0.0277	0.0798	<u>-0.0307</u>
rc1.4.8.fsm_A	-0.0279	0.0921	-0.0358	-0.0267	0.1056	<u>-0.0324</u>
rc1.4.8.fsm_C	-0.0403	0.0830	-0.0418	-0.0374	0.0956	<u>-0.0384</u>
rc1.4.9.fsm_A	-0.0284	0.0687	-0.0338	-0.0246	0.0862	<u>-0.0316</u>
rc1.4.9.fsm_C	-0.0402	0.0760	-0.0429	-0.0372	0.0863	<u>-0.0384</u>
rc1.6.10.fsm_A	-0.0234	0.0911	-0.0283	<u>-0.0196</u>	0.1014	-0.0165
rc1.6.10.fsm_C	-0.0315	0.0930	-0.0371	-0.0245	0.1027	<u>-0.0301</u>
rc1.6.1.fsm_A	-0.0166	0.0310	0.0107	<u>-0.0077</u>	0.0376	0.0157
rc1.6.1.fsm_C	-0.0361	0.0126	-0.0428	-0.0305	0.0267	<u>-0.0348</u>
rc1.6.2.fsm_A	-0.0286	0.0846	-0.0345	<u>-0.0242</u>	0.0920	-0.0044
rc1.6.2.fsm_C	-0.048	0.0711	-0.0433	<u>-0.0413</u>	0.0873	-0.0370
rc1.6.3.fsm_A	-0.0219	0.0841	-0.0217	<u>-0.0168</u>	0.0970	-0.0096
rc1.6.3.fsm_C	-0.0390	0.0962	-0.0449	-0.0323	0.1162	<u>-0.0356</u>
rc1.6.4.fsm_A	-0.0133	0.0639	-0.021	-0.0091	0.0811	<u>-0.0183</u>
rc1.6.4.fsm_C	-0.0312	0.0755	-0.0371	-0.0238	0.0952	<u>-0.0328</u>
rc1.6.5.fsm_A	-0.0238	0.0503	0.0034	<u>-0.0154</u>	0.0616	0.0097
rc1.6.5.fsm_C	-0.0399	0.0418	-0.0375	<u>-0.0243</u>	0.0514	-0.0218
rc1.6.6.fsm_A	-0.0212	0.0711	0.0018	<u>-0.0131</u>	0.0786	0.0197
rc1.6.6.fsm_C	-0.0306	0.0656	-0.0271	<u>-0.0278</u>	0.0835	-0.0100
rc1.6.7.fsm_A	-0.0185	0.0862	0.0056	<u>-0.0128</u>	0.0974	0.0238
rc1.6.7.fsm_C	-0.0427	0.0602	-0.0407	<u>-0.0352</u>	0.0810	-0.0276
rc1.6.8.fsm_A	-0.0222	0.0906	-0.0255	<u>-0.0134</u>	0.0977	-0.0128
rc1.6.8.fsm_C	-0.0394	0.0944	-0.0454	<u>-0.0355</u>	0.1082	-0.0347
rc1.6.9.fsm_A	-0.0226	0.0967	-0.0266	-0.0156	0.1039	<u>-0.0169</u>
rc1.6.9.fsm_C	-0.0405	0.0938	-0.0468	-0.0376	0.1122	<u>-0.0405</u>
rc1.8.10.fsm_A	-0.0281	0.0858	-0.0311	-0.0196	0.0938	<u>-0.0212</u>
rc1.8.10.fsm_C	-0.0509	0.0660	-0.055	-0.0437	0.0811	<u>-0.0454</u>
rc1.8.1.fsm_A	-0.0185	0.0447	-0.0007	<u>-0.0133</u>	0.0562	0.0166
rc1.8.1.fsm_C	0.1094	0.2121	0.1059	0.1225	0.2228	<u>0.1217</u>
rc1.8.2.fsm_A	-0.0308	0.0555	-0.0146	<u>-0.0237</u>	0.0733	0.0067
rc1.8.2.fsm_C	-0.1295	-0.0538	-0.1382	-0.1250	-0.0264	<u>-0.1261</u>
rc1.8.3.fsm_A	-0.0205	0.0730	-0.0056	<u>-0.0103</u>	0.0869	0.0048
rc1.8.3.fsm_C	-0.097	0.0132	-0.0969	-0.0878	0.0315	<u>-0.0923</u>
rc1.8.4.fsm_A	-0.0149	0.0647	-0.023	-0.0125	0.0847	<u>-0.021</u>
rc1.8.4.fsm_C	-0.1025	0.0024	-0.1020	-0.0973	0.0126	<u>-0.0985</u>
rc1.8.5.fsm_A	-0.0302	0.0635	-0.0099	<u>-0.0243</u>	0.0714	-0.0005
rc1.8.5.fsm_C	0.152	0.2631	0.1697	<u>0.1622</u>	0.2799	0.1809
rc1.8.6.fsm_A	-0.0424	0.0740	0.0037	<u>-0.0294</u>	0.0852	0.0127
rc1.8.6.fsm_C	-0.0655	0.0720	-0.0504	<u>-0.0557</u>	0.0787	-0.0405
rc1.8.7.fsm_A	-0.0297	0.0864	-0.0201	<u>-0.0237</u>	0.0982	0.0048
rc1.8.7.fsm_C	-0.0709	0.0589	-0.0531	<u>-0.0624</u>	0.0658	-0.0395
rc1.8.8.fsm_A	-0.0184	0.0989	-0.0147	<u>-0.0149</u>	0.1040	0.0095
rc1.8.8.fsm_C	-0.0672	0.0607	-0.0722	-0.0595	0.0706	<u>-0.0633</u>
rc1.8.9.fsm_A	-0.0305	0.0958	-0.0150	<u>-0.0217</u>	0.0983	0.0089
rc1.8.9.fsm_C	-0.0424	0.0955	-0.0476	<u>-0.0356</u>	0.1025	-0.0350

Table 2: (continued)

	Minimum Error Gap			Mean Error Gap		
	DRSCI	GSPI	PyVRP	DRSCI	GSPI	PyVRP
rc2_10_10_fsm_A	-0.1168	-0.0679	-0.1245	-0.0996	-0.0621	<u>-0.1007</u>
rc2_10_10_fsm_C	-0.0944	-0.0927	-0.113	-0.0786	-0.0829	<u>-0.1008</u>
rc2_10_1_fsm_A	-0.1152	-0.1068	-0.1429	-0.1052	-0.1001	<u>-0.1372</u>
rc2_10_1_fsm_C	-0.0587	-0.0702	-0.0748	-0.0526	-0.0653	<u>-0.0715</u>
rc2_10_2_fsm_A	-0.1158	-0.1180	-0.1363	-0.1070	-0.1100	<u>-0.1242</u>
rc2_10_2_fsm_C	-0.0645	-0.0801	-0.0846	-0.0608	-0.0743	<u>-0.0784</u>
rc2_10_3_fsm_A	-0.0854	-0.0616	-0.1039	-0.0743	-0.0495	<u>-0.0921</u>
rc2_10_3_fsm_C	-0.0666	-0.0780	-0.0857	-0.0640	-0.0740	<u>-0.0822</u>
rc2_10_4_fsm_A	-0.1089	-0.0516	-0.1042	-0.0986	-0.0294	<u>-0.1038</u>
rc2_10_4_fsm_C	-0.0922	-0.0551	-0.1059	-0.0784	-0.0394	<u>-0.0982</u>
rc2_10_5_fsm_A	-0.1529	-0.1604	-0.1533	-0.1469	<u>-0.1566</u>	-0.1468
rc2_10_5_fsm_C	-0.0761	-0.0865	-0.0923	-0.0704	-0.0847	<u>-0.0904</u>
rc2_10_6_fsm_A	-0.1414	-0.1631	-0.1416	-0.1293	<u>-0.15</u>	-0.1246
rc2_10_6_fsm_C	-0.0789	-0.0858	-0.0908	-0.0739	-0.0805	<u>-0.0856</u>
rc2_10_7_fsm_A	-0.1570	-0.1693	-0.174	-0.1405	<u>-0.165</u>	-0.1507
rc2_10_7_fsm_C	-0.0868	-0.1033	-0.1065	-0.0798	-0.0987	-0.1018
rc2_10_8_fsm_A	-0.1642	-0.1645	-0.1696	<u>-0.1542</u>	-0.1521	-0.1488
rc2_10_8_fsm_C	-0.0857	-0.1008	-0.1146	-0.0795	-0.0948	<u>-0.1117</u>
rc2_10_9_fsm_A	-0.1147	-0.0866	-0.1176	<u>-0.1019</u>	-0.0796	-0.0932
rc2_10_9_fsm_C	-0.0834	-0.0889	-0.1041	-0.0788	-0.0847	<u>-0.098</u>
rc2_2_10_fsm_A	-0.0901	-0.0890	-0.1007	-0.0855	-0.0850	<u>-0.1005</u>
rc2_2_10_fsm_C	-0.0161	-0.0268	-0.0244	-0.0112	<u>-0.016</u>	-0.0139
rc2_2_1_fsm_A	-0.0594	0.0125	-0.0821	-0.0541	0.0303	<u>-0.0686</u>
rc2_2_1_fsm_C	-0.0083	0.0352	-0.02	-0.0039	0.0385	<u>-0.018</u>
rc2_2_2_fsm_A	-0.0731	0.0076	-0.1032	-0.0632	0.0224	<u>-0.0984</u>
rc2_2_2_fsm_C	-0.0321	-0.0074	-0.0376	-0.0249	0.0005	<u>-0.0323</u>
rc2_2_3_fsm_A	-0.0372	0.0309	-0.0439	-0.0288	0.0378	<u>-0.0398</u>
rc2_2_3_fsm_C	-0.0364	-0.0157	-0.0472	-0.0300	0.0001	<u>-0.0432</u>
rc2_2_4_fsm_A	-0.0181	-0.0091	-0.0146	<u>-0.0016</u>	0.0272	-0.0016
rc2_2_4_fsm_C	-0.0308	0.0023	-0.0358	-0.0251	0.0110	<u>-0.0346</u>
rc2_2_5_fsm_A	-0.0717	-0.0446	-0.1112	-0.0587	-0.0172	<u>-0.0904</u>
rc2_2_5_fsm_C	-0.0281	-0.0124	-0.0419	-0.0238	0.0135	<u>-0.0311</u>
rc2_2_6_fsm_A	-0.0556	-0.0338	-0.1043	-0.0480	-0.0017	<u>-0.0779</u>
rc2_2_6_fsm_C	-0.0369	-0.0242	-0.0388	-0.0283	-0.0236	<u>-0.0335</u>
rc2_2_7_fsm_A	-0.0485	-0.0056	-0.0704	-0.0412	0.0002	<u>-0.0593</u>
rc2_2_7_fsm_C	-0.0446	-0.0153	-0.0498	-0.0268	-0.0098	<u>-0.04</u>
rc2_2_8_fsm_A	-0.0563	0.0009	-0.0731	-0.0382	0.0115	<u>-0.0603</u>
rc2_2_8_fsm_C	-0.0344	-0.0238	-0.0516	-0.0266	-0.0222	<u>-0.0417</u>
rc2_2_9_fsm_A	-0.0725	0.0190	-0.0956	-0.0414	0.0312	<u>-0.0711</u>
rc2_2_9_fsm_C	-0.0470	-0.0377	-0.0559	-0.0422	-0.0301	<u>-0.0502</u>
rc2_4_10_fsm_A	-0.0967	-0.0848	-0.1242	-0.0853	-0.0541	<u>-0.1116</u>
rc2_4_10_fsm_C	-0.0596	-0.0683	-0.0788	-0.0468	-0.0523	<u>-0.0708</u>
rc2_4_1_fsm_A	-0.0630	-0.0041	-0.1133	-0.0530	0.0082	<u>-0.0934</u>
rc2_4_1_fsm_C	-0.0334	-0.0051	-0.0377	-0.0264	-0.0019	<u>-0.028</u>
rc2_4_2_fsm_A	-0.0585	0.0404	-0.0883	-0.0436	0.0531	<u>-0.0655</u>
rc2_4_2_fsm_C	-0.0309	-0.0113	-0.0368	-0.0178	-0.0016	<u>-0.0339</u>
rc2_4_3_fsm_A	-0.0736	-0.0126	-0.1055	-0.0564	0.0161	<u>-0.0866</u>
rc2_4_3_fsm_C	-0.0648	-0.0454	-0.0709	-0.0565	-0.0398	<u>-0.0674</u>
rc2_4_4_fsm_A	-0.0422	0.0212	-0.0584	-0.0225	0.0349	<u>-0.0483</u>
rc2_4_4_fsm_C	-0.0648	-0.0513	-0.0677	-0.0589	-0.0200	<u>-0.0655</u>
rc2_4_5_fsm_A	-0.1047	-0.0358	-0.1465	-0.0950	-0.0250	<u>-0.1314</u>
rc2_4_5_fsm_C	-0.0557	-0.0447	-0.0661	-0.0484	-0.0403	<u>-0.0605</u>
rc2_4_6_fsm_A	-0.0714	-0.0405	-0.115	-0.0662	-0.0221	<u>-0.1035</u>
rc2_4_6_fsm_C	-0.0527	-0.0493	-0.0611	-0.0476	-0.0393	<u>-0.0553</u>
rc2_4_7_fsm_A	-0.0984	-0.0838	-0.1621	-0.0859	-0.0583	<u>-0.1409</u>
rc2_4_7_fsm_C	-0.0377	-0.0562	-0.0612	-0.0324	-0.0469	<u>-0.0572</u>
rc2_4_8_fsm_A	-0.1008	-0.0720	-0.1578	-0.0853	-0.0623	<u>-0.1283</u>
rc2_4_8_fsm_C	-0.0559	-0.0575	-0.071	-0.0513	-0.0528	<u>-0.0645</u>
rc2_4_9_fsm_A	-0.0790	-0.0726	-0.11	-0.0734	-0.0344	<u>-0.0932</u>
rc2_4_9_fsm_C	-0.0709	-0.0908	-0.0975	-0.0596	-0.0729	<u>-0.0866</u>
rc2_6_10_fsm_A	-0.0901	-0.0595	-0.0955	-0.0831	-0.0354	<u>-0.0866</u>
rc2_6_10_fsm_C	-0.0875	-0.0889	-0.1011	-0.0740	-0.0844	<u>-0.095</u>
rc2_6_1_fsm_A	-0.0935	-0.0705	-0.1262	-0.0825	-0.0581	<u>-0.1068</u>
rc2_6_1_fsm_C	-0.0406	-0.0433	-0.047	-0.0370	<u>-0.0389</u>	-0.0379
rc2_6_2_fsm_A	-0.1057	-0.0879	-0.1518	-0.1015	-0.0661	<u>-0.1421</u>
rc2_6_2_fsm_C	-0.0337	-0.0428	-0.0505	-0.0281	-0.0325	<u>-0.0451</u>
rc2_6_3_fsm_A	-0.1087	-0.0634	-0.1328	-0.0964	-0.0491	<u>-0.1291</u>
rc2_6_3_fsm_C	-0.0475	-0.0517	-0.0678	-0.0405	-0.0454	<u>-0.0612</u>
rc2_6_4_fsm_A	-0.0968	-0.0373	-0.1096	-0.0766	0.0024	<u>-0.1023</u>
rc2_6_4_fsm_C	-0.0642	-0.0227	-0.0801	-0.0560	-0.0126	<u>-0.0686</u>
rc2_6_5_fsm_A	-0.0940	-0.0964	-0.1347	-0.0908	-0.0773	<u>-0.1189</u>
rc2_6_5_fsm_C	-0.0484	-0.0486	-0.0599	-0.0440	-0.0442	<u>-0.0551</u>

Table 2: (continued)

	Minimum Error Gap			Mean Error Gap		
	DRSCI	GSPI	PyVRP	DRSCI	GSPI	PyVRP
rc2.6.6_fsm_A	-0.1368	-0.1492	-0.1761	-0.1298	-0.1472	<u>-0.1586</u>
rc2.6.6_fsm_C	-0.0652	-0.0703	-0.0776	-0.0594	-0.0646	<u>-0.0725</u>
rc2.6.7_fsm_A	-0.1352	-0.1575	-0.1631	-0.1229	-0.1319	<u>-0.1424</u>
rc2.6.7_fsm_C	-0.0593	-0.0699	-0.082	-0.0496	-0.0664	<u>-0.0749</u>
rc2.6.8_fsm_A	-0.1403	-0.1436	-0.1316	-0.1166	-0.1169	<u>-0.1249</u>
rc2.6.8_fsm_C	-0.0666	-0.0918	-0.095	-0.0582	-0.0835	<u>-0.0871</u>
rc2.6.9_fsm_A	-0.1052	-0.0721	-0.1288	<u>-0.0991</u>	-0.0648	<u>-0.0974</u>
rc2.6.9_fsm_C	-0.0716	-0.093	-0.0923	-0.0632	-0.0821	<u>-0.0863</u>
rc2.8.10_fsm_A	-0.1117	-0.0795	-0.1598	-0.1066	-0.0639	<u>-0.1271</u>
rc2.8.10_fsm_C	-0.0835	-0.0955	-0.1136	-0.0748	-0.0871	<u>-0.1053</u>
rc2.8.1_fsm_A	-0.0948	-0.0666	-0.1169	-0.0831	-0.0606	<u>-0.1157</u>
rc2.8.1_fsm_C	-0.0499	-0.0537	-0.06	-0.0428	-0.0495	<u>-0.0524</u>
rc2.8.2_fsm_A	-0.1092	-0.0656	-0.1242	-0.0911	-0.0579	<u>-0.1075</u>
rc2.8.2_fsm_C	-0.0515	-0.0515	-0.0616	-0.0464	-0.0451	<u>-0.0507</u>
rc2.8.3_fsm_A	-0.1047	-0.0864	-0.1434	-0.0936	-0.0731	<u>-0.1335</u>
rc2.8.3_fsm_C	-0.0588	-0.0708	-0.0742	-0.0551	-0.0642	<u>-0.0707</u>
rc2.8.4_fsm_A	-0.0966	-0.0512	-0.1158	-0.0782	-0.0276	<u>-0.1044</u>
rc2.8.4_fsm_C	-0.0673	-0.0281	-0.0854	-0.0589	-0.0206	<u>-0.0779</u>
rc2.8.5_fsm_A	-0.1319	-0.1099	-0.1456	-0.1149	-0.1048	<u>-0.138</u>
rc2.8.5_fsm_C	-0.0486	-0.0586	-0.0619	-0.0452	-0.0548	<u>-0.0581</u>
rc2.8.6_fsm_A	-0.1110	-0.1231	-0.1224	-0.1017	<u>-0.1152</u>	-0.1100
rc2.8.6_fsm_C	-0.0675	-0.0787	-0.0806	-0.0631	-0.0760	<u>-0.0773</u>
rc2.8.7_fsm_A	-0.1263	-0.1332	-0.1685	-0.1129	-0.1213	<u>-0.1464</u>
rc2.8.7_fsm_C	-0.0744	-0.0855	-0.0878	-0.0652	-0.0809	<u>-0.0853</u>
rc2.8.8_fsm_A	-0.1285	-0.1246	-0.1588	-0.1158	-0.1055	<u>-0.1205</u>
rc2.8.8_fsm_C	-0.0687	-0.0828	-0.0893	-0.0646	-0.0741	<u>-0.0865</u>
rc2.8.9_fsm_A	-0.1330	-0.1080	-0.1419	-0.1121	-0.0979	<u>-0.1298</u>
rc2.8.9_fsm_C	-0.0611	-0.0833	-0.0946	-0.0588	-0.0721	<u>-0.0878</u>

Table 3: Detailed results new HFVRPTW dataset: Minimum and average total costs per instance and solution method. Bolt marks the best result for each instance, and underlined entries denote the best average performance. Notation details: The instance name is divided into segments, each separated by “+”. The first segment references the underlying base instance, following the naming convention of [Gehring and Homberger \(1999\)](#). The second segments indicates the number of available vehicle types in the heterogeneous fleet, and the third segment denotes the ratio between fixed and variable costs. Segments 4 and 5 store the values for γ and $\sum_{m \in \mathcal{M} \setminus \underline{m}} \kappa_m$ respectively. Benchmark: PyVRP by [Wouda et al. \(2024\)](#).

	Minimum Total Costs		Mean Total Costs	
	DRSCI	PyVRP	DRSCI	PyVRP
c1_10.1+vt3+fed-2+ccr-0.8+lva-0.67	137517.2066	141170.0037	<u>137955.8523</u>	141663.7313
c1_10.1+vt3+fed-2+ccr-0.8+lva-0.9	130132.7188	137345.5138	<u>130188.1851</u>	137849.0523
c1_10.1+vt3+vcd-2+ccr-0.8+lva-0.67	73409.7962	75692.3375	<u>73524.9454</u>	78256.2623
c1_10.1+vt3+vcd-2+ccr-0.8+lva-0.9	67806.914	70588.3326	<u>67993.9563</u>	74236.7070
c1_10.1+vt5+fed-2+ccr-0.8+lva-0.67	141336.7918	145603.6274	<u>141637.7058</u>	146676.6596
c1_10.1+vt5+fed-2+ccr-0.8+lva-0.9	134077.529	143117.5863	<u>134393.4417</u>	143192.2731
c1_10.1+vt5+vcd-2+ccr-0.8+lva-0.67	75605.6978	80965.2095	<u>75714.6026</u>	81953.2455
c1_10.1+vt5+vcd-2+ccr-0.8+lva-0.9	70473.8192	76012.9262	<u>70614.6373</u>	80198.8954
c1_10.4+vt3+fed-2+ccr-0.8+lva-0.67	138512.0057	139040.3040	<u>138772.8156</u>	139906.9791
c1_10.4+vt3+fed-2+ccr-0.8+lva-0.9	130733.5497	130033.345	<u>130852.6625</u>	<u>130560.8827</u>
c1_10.4+vt3+vcd-2+ccr-0.8+lva-0.67	72619.6335	73415.6453	<u>72850.3927</u>	73577.1500
c1_10.4+vt3+vcd-2+ccr-0.8+lva-0.9	66883.7800	66626.1666	67032.1793	66873.4815
c1_10.4+vt5+fed-2+ccr-0.8+lva-0.67	144774.1401	145180.5297	<u>144933.2957</u>	145980.4142
c1_10.4+vt5+fed-2+ccr-0.8+lva-0.9	137075.0893	137648.7977	<u>137603.0023</u>	138287.6585
c1_10.4+vt5+vcd-2+ccr-0.8+lva-0.67	76236.0006	77119.3811	<u>76579.0868</u>	77311.9276
c1_10.4+vt5+vcd-2+ccr-0.8+lva-0.9	71128.3327	71471.0924	<u>71262.3088</u>	72262.3156
c2_10.1+vt3+fed-2+ccr-0.8+lva-0.67	56660.1202	54207.6441	56804.1669	<u>55589.1967</u>
c2_10.1+vt3+fed-2+ccr-0.8+lva-0.9	54221.6927	52748.8803	<u>54598.7272</u>	55471.3194
c2_10.1+vt3+vcd-2+ccr-0.8+lva-0.67	29123.7371	28492.049	29182.1599	28735.2978
c2_10.1+vt3+vcd-2+ccr-0.8+lva-0.9	27787.0985	27189.2858	27820.3025	<u>27495.5706</u>
c2_10.1+vt5+fed-2+ccr-0.8+lva-0.67	56563.6838	55866.0588	<u>56791.6005</u>	57838.7162
c2_10.1+vt5+vcd-2+ccr-0.8+lva-0.9	54428.2043	55267.3295	<u>54629.6278</u>	56706.9088
c2_10.1+vt5+vcd-2+ccr-0.8+lva-0.67	29736.3514	28801.5045	29856.4220	<u>28942.2436</u>
c2_10.1+vt5+vcd-2+ccr-0.8+lva-0.9	28091.1393	27390.1889	28116.3637	28188.6701
c2_10.4+vt3+fed-2+ccr-0.8+lva-0.67	51590.3317	49680.5448	51694.9041	<u>49916.792</u>
c2_10.4+vt3+fed-2+ccr-0.8+lva-0.9	49425.6460	46908.9056	49592.8965	<u>47051.821</u>
c2_10.4+vt3+vcd-2+ccr-0.8+lva-0.67	27087.4241	25592.8871	27105.0048	<u>25754.1514</u>
c2_10.4+vt3+vcd-2+ccr-0.8+lva-0.9	25628.8170	24171.544	25794.6147	<u>24444.6935</u>
c2_10.4+vt5+fed-2+ccr-0.8+lva-0.67	52805.6088	50456.002	53319.5130	<u>50617.9253</u>
c2_10.4+vt5+fed-2+ccr-0.8+lva-0.9	50142.0909	47812.4153	50891.3453	<u>48388.841</u>
c2_10.4+vt5+vcd-2+ccr-0.8+lva-0.67	27807.0460	26355.0567	27935.4069	<u>26443.6319</u>
c2_10.4+vt5+vcd-2+ccr-0.8+lva-0.9	26025.7614	24720.4376	26111.6379	<u>24957.9815</u>
r1_10.1+vt3+fed-2+ccr-0.8+lva-0.67	186787.8751	179487.5664	188131.6998	<u>179913.557</u>
r1_10.1+vt3+fed-2+ccr-0.8+lva-0.9	179123.1465	173412.9686	180828.3801	<u>175707.1146</u>
r1_10.1+vt3+vcd-2+ccr-0.8+lva-0.67	96243.3785	95293.2333	<u>96857.8355</u>	97422.9765
r1_10.1+vt3+vcd-2+ccr-0.8+lva-0.9	90597.8468	96074.4945	<u>91528.4633</u>	96374.9401
r1_10.1+vt5+vcd-2+ccr-0.8+lva-0.67	188617.7960	182864.3596	190508.5405	<u>183328.5411</u>
r1_10.1+vt5+vcd-2+ccr-0.8+lva-0.9	183049.7307	175402.5769	183803.2080	<u>176251.8907</u>
r1_10.1+vt5+vcd-2+ccr-0.8+lva-0.67	98739.1577	98106.1717	99895.5714	<u>99300.6382</u>
r1_10.1+vt5+vcd-2+ccr-0.8+lva-0.9	93829.6801	95676.0882	<u>95114.687</u>	96898.0197
r1_10.4+vt3+fed-2+ccr-0.8+lva-0.67	152095.4020	150622.2034	152193.6814	<u>150795.7059</u>
r1_10.4+vt3+fed-2+ccr-0.8+lva-0.9	143061.3253	140793.5326	143281.1625	<u>141229.1842</u>
r1_10.4+vt3+vcd-2+ccr-0.8+lva-0.67	79177.1636	78380.3202	79517.3381	<u>78677.5319</u>
r1_10.4+vt3+vcd-2+ccr-0.8+lva-0.9	73964.7603	72497.0084	74266.8923	<u>72782.5667</u>
r1_10.4+vt5+fed-2+ccr-0.8+lva-0.67	155804.8222	154142.8792	155960.4954	<u>154378.0799</u>
r1_10.4+vt5+fed-2+ccr-0.8+lva-0.9	147354.2543	145632.4178	147598.7496	<u>145996.9367</u>
r1_10.4+vt5+vcd-2+ccr-0.8+lva-0.67	82412.6197	81534.5673	82784.2868	<u>81843.2918</u>
r1_10.4+vt5+vcd-2+ccr-0.8+lva-0.9	76728.6024	75692.913	77056.5028	<u>76053.169</u>
r2_10.1+vt3+fed-2+ccr-0.8+lva-0.67	86124.7993	82127.2906	86761.1157	<u>82422.6019</u>
r2_10.1+vt3+fed-2+ccr-0.8+lva-0.9	86169.4489	82907.6428	86561.7451	<u>83237.2497</u>
r2_10.1+vt3+vcd-2+ccr-0.8+lva-0.67	51393.3710	49021.4354	51883.9848	<u>49105.9295</u>
r2_10.1+vt3+vcd-2+ccr-0.8+lva-0.9	51617.3342	48893.5766	52274.9294	<u>48975.8772</u>
r2_10.1+vt5+fed-2+ccr-0.8+lva-0.67	86297.2009	81975.6034	87032.5177	<u>82575.5945</u>
r2_10.1+vt5+fed-2+ccr-0.8+lva-0.9	85257.2830	82102.8751	86559.7493	<u>82409.4556</u>
r2_10.1+vt5+vcd-2+ccr-0.8+lva-0.67	51800.9819	49015.6927	52388.4844	<u>49076.6773</u>
r2_10.1+vt5+vcd-2+ccr-0.8+lva-0.9	51602.8814	48878.361	52171.5829	<u>49081.7221</u>
r2_10.4+vt3+fed-2+ccr-0.8+lva-0.67	59746.6000	57016.0212	59861.6360	<u>57146.9587</u>
r2_10.4+vt3+fed-2+ccr-0.8+lva-0.9	57737.1353	54934.7831	58519.7691	<u>55074.796</u>
r2_10.4+vt3+vcd-2+ccr-0.8+lva-0.67	30196.6103	28415.5097	30441.6465	<u>28484.1673</u>
r2_10.4+vt3+vcd-2+ccr-0.8+lva-0.9	29257.4616	27517.1528	29287.2487	<u>27580.4773</u>
r2_10.4+vt5+fed-2+ccr-0.8+lva-0.67	60774.4599	57676.8915	60935.4240	<u>57717.1234</u>
r2_10.4+vt5+fed-2+ccr-0.8+lva-0.9	58842.9028	55551.2496	59822.0612	<u>55674.5378</u>
r2_10.4+vt5+vcd-2+ccr-0.8+lva-0.67	30935.8146	28705.6977	31143.0586	<u>28853.7759</u>
r2_10.4+vt5+vcd-2+ccr-0.8+lva-0.9	30061.4072	27868.4996	30161.2000	<u>28127.5491</u>
rc1_10.1+vt3+fed-2+ccr-0.8+lva-0.67	163166.4683	161117.0641	163927.4056	<u>161817.5564</u>
rc1_10.1+vt3+fed-2+ccr-0.8+lva-0.9	153002.7710	152277.0256	153376.7884	<u>153127.4627</u>

Table 3: (continued)

	Minimum Total Costs		Mean Total Costs	
	DRSCI	PyVRP	DRSCI	PyVRP
rc1_10.1+vt3+vcd-2+ccr-0.8+lva-0.67	84360.8054	83404.8858	84546.0943	<u>84317.8954</u>
rc1_10.1+vt3+vcd-2+ccr-0.8+lva-0.9	78789.7908	78487.6881	<u>78958.1608</u>	80523.5357
rc1_10.1+vt5+fcd-2+ccr-0.8+lva-0.67	166221.3626	165155.235	167115.3974	<u>165498.4496</u>
rc1_10.1+vt5+fcd-2+ccr-0.8+lva-0.9	157555.7869	156316.1984	158168.1181	<u>157332.0958</u>
rc1_10.1+vt5+vcd-2+ccr-0.8+lva-0.67	86873.3457	87730.1871	<u>87060.0355</u>	88362.4997
rc1_10.1+vt5+vcd-2+ccr-0.8+lva-0.9	81046.8779	82571.5934	<u>81353.2632</u>	83967.3815
rc1_10.4+vt3+fcd-2+ccr-0.8+lva-0.67	147433.9664	146120.5168	147750.3740	<u>146275.2175</u>
rc1_10.4+vt3+fcd-2+ccr-0.8+lva-0.9	138699.1355	136647.2525	138877.5790	<u>137027.3218</u>
rc1_10.4+vt3+vcd-2+ccr-0.8+lva-0.67	76910.2720	76457.8368	77284.1196	<u>76703.1516</u>
rc1_10.4+vt3+vcd-2+ccr-0.8+lva-0.9	71021.9856	69785.1054	71459.1735	<u>70116.1465</u>
rc1_10.4+vt5+fcd-2+ccr-0.8+lva-0.67	150627.3236	149016.9824	151140.9246	<u>149550.1508</u>
rc1_10.4+vt5+fcd-2+ccr-0.8+lva-0.9	142145.7149	141034.4903	142628.5762	<u>141713.1527</u>
rc1_10.4+vt5+vcd-2+ccr-0.8+lva-0.67	79874.0795	78736.4961	80343.3902	<u>79100.5278</u>
rc1_10.4+vt5+vcd-2+ccr-0.8+lva-0.9	74227.8748	72888.3944	74428.0852	<u>73520.6223</u>
rc2_10.1+vt3+fcd-2+ccr-0.8+lva-0.67	75404.5610	69822.3761	76126.6562	<u>71215.0143</u>
rc2_10.1+vt3+fcd-2+ccr-0.8+lva-0.9	73560.9401	72150.4476	73615.1688	<u>72495.6539</u>
rc2_10.1+vt3+vcd-2+ccr-0.8+lva-0.67	41232.9601	39514.7508	41558.3558	<u>39873.0473</u>
rc2_10.1+vt3+vcd-2+ccr-0.8+lva-0.9	41040.6265	40156.4984	41094.2788	<u>40198.8219</u>
rc2_10.1+vt5+fcd-2+ccr-0.8+lva-0.67	74117.9294	71064.7805	74856.3391	<u>72170.2847</u>
rc2_10.1+vt5+fcd-2+ccr-0.8+lva-0.9	73693.3218	68882.5392	73867.3517	<u>70558.2142</u>
rc2_10.1+vt5+vcd-2+ccr-0.8+lva-0.67	40828.3045	39850.4073	41089.5619	<u>39998.9792</u>
rc2_10.1+vt5+vcd-2+ccr-0.8+lva-0.9	40823.0178	38998.0988	41190.7715	<u>39669.5202</u>
rc2_10.4+vt3+fcd-2+ccr-0.8+lva-0.67	53107.8292	51804.6645	53250.5589	<u>51921.2216</u>
rc2_10.4+vt3+fcd-2+ccr-0.8+lva-0.9	50923.4238	49787.9043	51380.8873	<u>49854.177</u>
rc2_10.4+vt3+vcd-2+ccr-0.8+lva-0.67	26906.7365	25739.0622	27074.1794	<u>25765.1488</u>
rc2_10.4+vt3+vcd-2+ccr-0.8+lva-0.9	25885.3244	24718.5461	25938.0465	<u>24780.4153</u>
rc2_10.4+vt5+fcd-2+ccr-0.8+lva-0.67	54261.9231	51534.3176	54932.5515	<u>51611.5726</u>
rc2_10.4+vt5+fcd-2+ccr-0.8+lva-0.9	52551.6101	50517.331	53116.5865	<u>50587.3615</u>
rc2_10.4+vt5+vcd-2+ccr-0.8+lva-0.67	27784.2569	25755.5997	27830.5745	<u>25852.055</u>
rc2_10.4+vt5+vcd-2+ccr-0.8+lva-0.9	26162.5870	25055.7288	26448.9435	<u>25116.758</u>

References

- Bräysy, O., Porkka, P.P., Dullaert, W., Repoussis, P.P., Tarantilis, C.D., 2009. A well-scalable metaheuristic for the fleet size and mix vehicle routing problem with time windows. *Expert Systems with Applications* 36, 8460–8475.
- Gehring, H., Homberger, J., 1999. A parallel hybrid evolutionary metaheuristic for the vehicle routing problem with time windows, in: Miettinen, K., Mäkelä, M., Toivanen, J. (Eds.), *Proc. EUROGEN99*, University of Jyväskylä, Jyväskylä, Finland. pp. 57–64.
- Máximo, V.R., Cordeau, J.F., Nascimento, M.C., 2022. An adaptive iterated local search heuristic for the heterogeneous fleet vehicle routing problem. *Computers & Operations Research* 148, 105954.
- Pessoa, A., Sadykov, R., Uchoa, E., 2018. Enhanced branch-cut-and-price algorithm for heterogeneous fleet vehicle routing problems. *European Journal of Operational Research* 270, 530–543.
- Wouda, N.A., Lan, L., Kool, W., 2024. PyVRP: A high-performance VRP solver package. *INFORMS Journal on Computing* 36, 943–955.